# Group & Ring Theory MATP33: Course Summary of Important Results

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## Lecture 0 — Background Results

**Theorem 1** (First Isomorphism Theorem)**.** *Let $f : G \to H$ be a group homomorphism. Then* $\ker f \trianglelefteq G$ *and* $\operatorname{im} f \cong G/\ker f$*.*

**Theorem 2** (Second Isomorphism Theorem)**.** *Let $G$ be a group, $H \leq G$, and $N \trianglelefteq G$. Then $HN \leq G$, $H \cap N \trianglelefteq H$, and $HN/N \cong H/(H \cap N)$.*

**Theorem 3** (Third Isomorphism Theorem)**.** *If $K \trianglelefteq G$ and $K \leq L \trianglelefteq G$, then $(G/K)/(L/K) \cong G/L$.*

**Theorem 4** (Correspondence Theorem)**.** *Let $N \trianglelefteq G$. Every subgroup of $G/N$ is of the form $H/N$ where $N \leq H \leq G$, and conversely if $N \leq H \leq G$ then $H/N \leq G/N$.*

**Theorem 5** (Lagrange)**.** *If $H \leq G$ and $G$ is finite, then $|H|$ divides $|G|$.*

**Theorem 6** (Fundamental Theorem of Finitely Generated Abelian Groups)**.** *Let $A$ be a finitely generated abelian group. Then*

$$A \cong \mathbb{Z}^s \oplus \mathbb{Z}/a_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/a_r\mathbb{Z},$$

*where $s \geq 0$ and $a_1 \mid a_2 \mid \cdots \mid a_r$ are non-zero non-units in $\mathbb{Z}$. This decomposition is unique.*

## Lecture 1

**Theorem 7** (1.10)**.** *Let $G$ be a group and $X$ a set. Then:*

*(i) If $X$ is a $G$-set, then the action of $G$ on $X$ induces a homomorphism $\varphi : G \to S_X$;*

*(ii) Any homomorphism $\varphi : G \to S_X$ induces an action of $G$ on $X$.*

*Proof.* (i) Define $\varphi : G \to S_X$ by $(\varphi(a))(x) = a \cdot x$ for $a \in G$ and $x \in X$. We must first check that $\varphi(a) \in S_X$ for each $a \in G$, i.e. that the function $x \mapsto a \cdot x$ is a bijection $X \to X$.

- *Injective:* Suppose $a \cdot x = a \cdot y$. Then $x = a^{-1} \cdot (a \cdot x) = a^{-1} \cdot (a \cdot y) = y$.
- *Surjective:* For any $x \in X$, we have $a \cdot (a^{-1} \cdot x) = x$, so $x \in \operatorname{im}(\varphi(a))$.

Hence $\varphi(a) \in S_X$. To see $\varphi$ is a homomorphism, let $a, b \in G$. Then for all $x \in X$,

$$(\varphi(ab))(x) = (ab) \cdot x = a \cdot (b \cdot x) = (\varphi(a))((\varphi(b))(x)) = (\varphi(a)\varphi(b))(x).$$

(ii) Define $a \cdot x := (\varphi(a))(x)$. Then $(ab) \cdot x = (\varphi(ab))(x) = (\varphi(a)\varphi(b))(x) = a \cdot (b \cdot x)$, and $e \cdot x = (\varphi(e))(x) = x$. Hence $X$ is a $G$-set. □

**Theorem 8** (1.11 — Cayley's Theorem)**.** *Every group $G$ is isomorphic to a subgroup of the symmetric group $S_G$.*

*Proof.* By Theorem 1.10(i), left translation $a \cdot x := ax$ gives a homomorphism $\varphi : G \to S_G$. If $a \in \ker \varphi$, then $\varphi(a) = \mathrm{id}_G$, so $ax = x$ for all $x \in G$, giving $a = e$. Hence $\ker \varphi = \{e\}$, and by the First Isomorphism Theorem $G \cong G/\ker \varphi \cong \operatorname{im} \varphi \leq S_G$. □

**Definition 1** (Normal core)**.** *Let $G$ be a group and $H \leq G$. The* normal core *of $H$ in $G$ is*

$$\mathrm{core}(H) := \bigcap_{x \in G} xHx^{-1}.$$

*It is the largest normal subgroup of $G$ contained in $H$.*

**Theorem 9** (1.12)**.** *Let $G$ be a group and $H \leq G$ a subgroup of index $n$. Then there is a homomorphism $\varphi : G \to S_n$ such that $\ker \varphi = \mathrm{core}(H)$.*

*Proof.* Since $|G/H| = n$, we have $S_{G/H} \cong S_n$. By Theorem 1.10(i), the action of $G$ on the coset space $G/H$ by left multiplication gives a homomorphism $\varphi : G \to S_n$ with $\varphi(a)(xH) = axH$. Then

$$\begin{aligned}
\ker \varphi &= \{a \in G \mid axH = xH \text{ for all } x \in G\} \\
&= \{a \in G \mid x^{-1}ax \in H \text{ for all } x \in G\} \\
&= \{a \in G \mid a \in xHx^{-1} \text{ for all } x \in G\} \\
&= \bigcap_{x \in G} xHx^{-1} = \mathrm{core}(H).
\end{aligned}$$

□

**Corollary 1** (1.13)**.** *Let $G$ be a group with a normal subgroup $H$ of index $n$. Then $G/H$ is isomorphic to a subgroup of $S_n$.*

*Proof.* Since $H \trianglelefteq G$, conjugation fixes $H$, so $\mathrm{core}(H) = \bigcap_{x \in G} xHx^{-1} = H$. By Theorem 1.12, $\ker \varphi = H$, and by the First Isomorphism Theorem $G/H \cong \operatorname{im} \varphi \leq S_n$. □

**Theorem 10** (1.24 — Orbit-Stabiliser Theorem)**.** *Let $G$ be a group acting on a set $X$. Then the set of all orbits in $X$ under $G$ is a partition of $X$. Furthermore, for each $x \in X$ there is a bijection $Gx \to G/G_x$, and hence*

$$|Gx| = |G : G_x|.$$

*Therefore, if $X$ is finite,*

$$|X| = \sum_{x \in C} |G : G_x|,$$

*where $C \subseteq X$ contains exactly one element from each orbit.*

*Proof. (Partition.)* Define $x \sim y$ if $x = ay$ for some $a \in G$. This is an equivalence relation: reflexivity follows from $e \cdot x = x$, symmetry from $x = ay \Rightarrow y = a^{-1}x$, and transitivity from $x = ay$, $y = bz \Rightarrow x = (ab)z$. The equivalence class of $x$ is exactly the orbit $Gx$, so the orbits partition $X$.

*(Bijection $Gx \to G/G_x$.)* Define

$$\varphi : Gx \to G/G_x, \qquad ax \mapsto aG_x.$$

The key observation is the following chain of equivalences, for any $a, b \in G$:

$$ax = bx \iff a^{-1}bx = x \iff a^{-1}b \in G_x \iff aG_x = bG_x.$$

- *Well-defined:* If $ax = bx$ (the same orbit element written using two different group elements), then $aG_x = bG_x$, so the coset $\varphi$ assigns does not depend on the choice of representative.
- *Injective:* If $\varphi(ax) = \varphi(bx)$, i.e. $aG_x = bG_x$, then $ax = bx$ by the chain above.
- *Surjective:* Take any coset $aG_x \in G/G_x$. Since $a \in G$, we have $ax \in Gx$, and $\varphi(ax) = aG_x$.

Hence $\varphi$ is a bijection, giving $|Gx| = |G/G_x| = |G : G_x|$. Summing over one representative per orbit gives $|X| = \sum_{x \in C} |G : G_x|$. □

**Theorem 11** (1.25 — Class Equation)**.** *Let $G$ be a finite group. Then the conjugacy classes partition $G$, and*

$$|G| = \sum_{a \in C} |G : C_G(a)|,$$

*where $C \subseteq G$ contains exactly one element from each conjugacy class.*

*Proof.* Let $G$ act on itself by conjugation, i.e. $a \cdot x = axa^{-1}$. The orbit of $x$ under this action is its conjugacy class $\mathrm{ccl}(x) = \{axa^{-1} \mid a \in G\}$, and the stabiliser of $x$ is $G_x = \{a \in G \mid axa^{-1} = x\} = C_G(x)$. Applying Theorem 1.24 directly gives the partition into orbits and $|G| = \sum_{a \in C} |G : C_G(a)|$. □

# Lecture 2

**Theorem 12** (2.1)**.** *Let $G$ be a finite group of order $p^n$, where $p$ is prime and $n \in \mathbb{N}$. Then:*

*(i) $G$ has a non-trivial centre $Z(G)$;*

*(ii) $Z(G) \cap K$ is non-trivial for any non-trivial normal subgroup $K$ of $G$;*

*(iii) if $H$ is a proper subgroup of $G$, then $H \subsetneq N_G(H)$. In particular, if $|H| = p^{n-1}$ then $H \trianglelefteq G$.*

*Proof.* (i) Since $x \in Z(G)$ iff $\mathrm{ccl}(x) = \{x\}$, the class equation refines to

$$p^n = |Z(G)| + \sum_{x \in C} |G : C_G(x)|,$$

where $C$ runs over conjugacy classes with more than one element. For each $x \in C$, $|C_G(x)|$ properly divides $p^n$, so $p \mid |G : C_G(x)|$. Hence $p$ divides the sum, and therefore $p \mid |Z(G)|$. Since $e \in Z(G)$, we have $|Z(G)| \geq 1$, so in fact $|Z(G)| \geq p > 1$.

(ii) Intersecting the class decomposition of $G$ with $K$ gives $|K| = |Z(G) \cap K| + \sum_{x \in C} |\operatorname{ccl}(x) \cap K|$. Since $K \trianglelefteq G$, each conjugacy class is either disjoint from $K$ or contained in $K$, so $p \mid \sum |\operatorname{ccl}(x) \cap K|$. As $p \mid |K|$, we get $p \mid |Z(G) \cap K|$, and the same argument as (i) gives $Z(G) \cap K$ non-trivial.

(iii) Let $M$ be a maximal normal subgroup of $G$ contained in $H$. Then $G/M$ is a p-group, so by (i) it has a non-trivial centre $L/M \trianglelefteq G/M$. By the Correspondence Theorem, $L \trianglelefteq G$. Since $M \subsetneq L$ strictly, maximality of $M$ forces $L \not\subseteq H$.

It remains to show $L \subseteq N_G(H)$. To show $\ell \in N_G(H)$ we need $\ell^{-1}h\ell \in H$ for all $h \in H$. Take $h \in H$ and $\ell \in L$. Since $\ell M \in L/M = Z(G/M)$, it commutes with every coset, so $(\ell M)(hM) = (hM)(\ell M)$, i.e. $\ell hM = h\ell M$. This gives $\ell^{-1}h\ell \in hM$. Since $h \in H$ and $M \subseteq H$, we have $hM \subseteq H$, so $\ell^{-1}h\ell \in H$. Hence $\ell \in N_G(H)$, so $L \subseteq N_G(H)$.

Since $L \subseteq N_G(H)$ but $L \not\subseteq H$, we have $H \subsetneq N_G(H)$. Finally, if $|H| = p^{n-1}$ then $N_G(H)$ has order strictly greater than $p^{n-1}$, and the only possibility is $p^n = |G|$, so $N_G(H) = G$ and $H \trianglelefteq G$. □

**Theorem 13** (2.2 — Burnside's Lemma)**.** *Let $G$ be a finite group acting on a finite set $X$, and for $g \in G$ let $X_g := \{x \in X \mid gx = x\}$. Then the number $k$ of orbits in $X$ under $G$ is*

$$k = \frac{1}{|G|} \sum_{g \in G} |X_g|.$$

*Proof.* Let $S := \{(g, x) \in G \times X \mid gx = x\}$. Fixing $g$ gives $|S| = \sum_{g \in G} |X_g|$; fixing $x$ gives $|S| = \sum_{x \in X} |G_x|$. Let $C \subseteq X$ contain one element from each orbit. By the Orbit-Stabiliser Theorem, $|G_x| = |G|/|Gx| = |G|/|Ga|$ for all $x \in Ga$. Hence

$$\frac{1}{|G|} \sum_{g \in G} |X_g| = \frac{|S|}{|G|} = \frac{1}{|G|} \sum_{x \in X} |G_x| = \frac{1}{|G|} \sum_{a \in C} \sum_{x \in Ga} \frac{|G|}{|Ga|} = \frac{1}{|G|} \sum_{a \in C} |G| = k. \qquad \square$$

**Remark 1** (Pólya counting)**.** *Burnside's Lemma counts distinct objects under symmetry. As an illustration: how many necklaces with $p$ beads ($p$ prime) can be made from $n$ colours, where two necklaces are the same if one is a rotation of the other?*

*Let $X$ be the set of all colourings, so $|X| = n^p$. The cyclic group $G = \langle a \rangle$ of order $p$ acts on $X$ by rotation. For the identity, $X_e = X$ so $|X_e| = n^p$. For any $g \neq e$: since $|G|$ is prime, $g$ generates $G$, so a necklace fixed by $g$ must be fixed by all rotations, meaning all beads have the same colour. Hence $|X_g| = n$. Burnside then gives*

$$k = \frac{1}{p}\left(n^p + (p-1)n\right).$$

# Lecture 3

**Definition 2** (p-group, Sylow p-subgroup)**.** *Let $p$ be a prime. A group $G$ is a* $p$-group *if every element of $G$ has order a power of $p$. If $G$ is finite and $p^m \mid |G|$ but $p^{m+1} \nmid |G|$, then any subgroup of $G$ of order $p^m$ is called a* Sylow $p$-subgroup *of $G$.*

**Theorem 14** (3.5 — First Sylow Theorem)**.** *Let $G$ be a finite group and $p$ a prime. If $p^m \mid |G|$, then $G$ has a subgroup of order $p^m$.*

*Proof.* If $m = 0$ the result is trivial. We proceed by induction on $n = |G|$, the case $n = 1$ being trivial. Assume the result holds for all groups of order less than $n$.

*Case 1:* $p \mid |Z(G)|$. By the Fundamental Theorem of Finitely Generated Abelian Groups, $Z(G)$ contains an element $a$ of order $p$. Let $C = \langle a \rangle$. Since $a \in Z(G)$, we have $C \trianglelefteq G$, and $|G/C| = n/p$, which is divisible by $p^{m-1}$. By the induction hypothesis, $G/C$ has a subgroup of order $p^{m-1}$, which by the Correspondence Theorem is of the form $H/C$ for some $H \leq G$. Then $|H| = |H/C| \cdot |C| = p^{m-1} \cdot p = p^m$.

*Case 2:* $p \nmid |Z(G)|$. From the class equation $n = |Z(G)| + \sum_a |G : C_G(a)|$, since $p \mid n$ but $p \nmid |Z(G)|$, there must exist some $a \notin Z(G)$ with $p \nmid |G : C_G(a)|$. Since $|G| = |G : C_G(a)| \cdot |C_G(a)|$ and $p^m \mid |G|$ but $p \nmid |G : C_G(a)|$, it follows that $p^m \mid |C_G(a)|$. Since $a \notin Z(G)$, we have $C_G(a) \neq G$, so $|C_G(a)| < n$. By the induction hypothesis $C_G(a)$ contains a subgroup of order $p^m$, which is also a subgroup of $G$. □

**Corollary 2** (3.6 — Cauchy's Theorem)**.** *If $p$ is a prime dividing $|G|$, then $G$ has an element of order $p$.*

*Proof.* By the First Sylow Theorem, $G$ has a subgroup $H$ of order $p$. Every non-identity element of $H$ has order $p$. □

**Corollary 3** (3.7)**.** *A finite group $G$ is a $p$-group if and only if $|G|$ is a power of $p$.*

*Proof.* If $|G|$ is a power of $p$, then by Lagrange every element order divides $|G|$, hence is a power of $p$. Conversely, if $|G|$ is not a power of $p$, then some prime $q \neq p$ divides $|G|$, and by Cauchy's Theorem $G$ has an element of order $q$, contradicting that $G$ is a $p$-group. □

**Theorem 15** (3.8 — Second Sylow Theorem)**.** *Let $G$ be a finite group and $p$ a prime. All Sylow $p$-subgroups of $G$ are conjugate.*

*Proof.* Let $S, H \in \mathrm{Syl}_p(G)$. Let $S$ act on the coset space $G/H$ by left translation: $s \cdot gH = sgH$.

*Orbits have $p$-power size.* By the Orbit-Stabiliser Theorem, each orbit has size $|S : S_{gH}|$, which divides $|S| = p^m$, hence is a power of $p$.

*There must be a fixed point.* Since $H$ is a Sylow $p$-subgroup, $p \nmid |G|/|H| = |G/H|$. Writing $|G/H|$ as a sum of orbit sizes, each of which is a power of $p$, the only way this sum is not divisible by $p$ is if at least one orbit has size $p^0 = 1$.

*A fixed point gives conjugacy.* If $gH$ is a fixed point, then $s \cdot gH = gH$ for all $s \in S$, so $g^{-1}sg \in H$ for all $s \in S$. Hence $g^{-1}Sg \subseteq H$, and since $|g^{-1}Sg| = |S| = |H|$, we get $g^{-1}Sg = H$. So $S$ and $H$ are conjugate. □

**Corollary 4** (3.10)**.** *A Sylow $p$-subgroup of $G$ is unique if and only if it is normal in $G$.*

*Proof.* If $P$ is the unique Sylow $p$-subgroup, then $gPg^{-1} = P$ for all $g \in G$ (since conjugates are also Sylow $p$-subgroups), so $P \trianglelefteq G$. Conversely, if $P \trianglelefteq G$ then $P$ is its own conjugate, so by the Second Sylow Theorem it is the only Sylow $p$-subgroup. □

**Theorem 16** (3.9 — Third Sylow Theorem)**.** *Let $G$ be a finite group and $p$ a prime. The number $n_p$ of Sylow $p$-subgroups satisfies $n_p \mid |G|$ and $n_p \equiv 1 \pmod{p}$.*

*Proof.* $n_p \mid |G|$. By the Second Sylow Theorem, all Sylow $p$-subgroups are conjugate to a fixed $K \in \mathrm{Syl}_p(G)$, so $n_p = |\mathrm{ccl}(K)| = |G : N_G(K)|$, which divides $|G|$.

$n_p \equiv 1 \pmod{p}$. Let $P \in \mathrm{Syl}_p(G)$. Since conjugates of Sylow $p$-subgroups are again Sylow $p$-subgroups (by the Second Sylow Theorem), $P$ acts on the set $\mathrm{Syl}_p(G)$ by conjugation:

$$x \cdot Q := xQx^{-1}, \quad x \in P,\ Q \in \mathrm{Syl}_p(G).$$

The orbit of $Q$ is $P \cdot Q = \{xQx^{-1} : x \in P\}$, and its stabiliser is $\{x \in P : xQx^{-1} = Q\} = P \cap N_G(Q) = N_P(Q)$. By the Orbit-Stabiliser Theorem,

$$|P \cdot Q| = |P : N_P(Q)|,$$

which divides $|P| = p^m$ and is therefore a power of $p$. In particular $|P \cdot Q| = 1$ if and only if $N_P(Q) = P$.

Let $F = \{Q \in \mathrm{Syl}_p(G) : N_P(Q) = P\}$ be the set of fixed points, and choose representatives $Q_1, \ldots, Q_r$, one from each orbit of size greater than 1. The orbits partition $\mathrm{Syl}_p(G)$, so

$$n_p = |\mathrm{Syl}_p(G)| = |F| + \sum_{i=1}^{r} |P : N_P(Q_i)|.$$

Each $|P : N_P(Q_i)|$ is a power $p^{e_i}$ with $e_i \geq 1$, hence divisible by $p$. Therefore $n_p \equiv |F| \pmod{p}$.

It remains to show $F = \{P\}$, which gives $n_p \equiv 1 \pmod{p}$. Clearly $P \in F$. If $Q \in F$, then $xQx^{-1} = Q$ for all $x \in P$, so $P \leq N_G(Q)$. By Corollary 3.10, since $Q \trianglelefteq N_G(Q)$, $Q$ is the unique Sylow $p$-subgroup of $N_G(Q)$. But $P$ is also a Sylow $p$-subgroup of $N_G(Q)$, so $P = Q$. Hence $F = \{P\}$. □

# Lecture 4 — Modules: the Basics

**Definition 3** (4.1 — Left $R$-module)**.** *Let $R$ be a ring. An abelian group $M$ is a* left $R$-module *if there is a map $R \times M \to M$, $(r, m) \mapsto rm$, such that for all $r, r_1, r_2 \in R$ and $m, m_1, m_2 \in M$:*

*(i)* $r(m_1 + m_2) = rm_1 + rm_2$;

*(ii)* $(r_1 + r_2)m = r_1m + r_2m$;

*(iii)* $(r_1r_2)m = r_1(r_2m)$;

*(iv)* $1m = m$ *if* $1 \in R$.

**Lemma 1** (4.7)**.** *Let $M$ be an $R$-module. Then for $a \in R$ and $m \in M$:*

*(i)* $0m = 0$;

*(ii)* $a0 = 0$;

*(iii)* $(-a)m = -(am) = a(-m)$.

*Proof.* (i) $0m = (0 + 0)m = 0m + 0m$. Subtracting $0m$ from both sides gives $0 = 0m$.

(ii) $a0 = a(0 + 0) = a0 + a0$. Subtracting $a0$ from both sides gives $0 = a0$.

(iii) By (i) and axiom (ii): $0 = 0m = (a+(-a))m = am+(-a)m$, so $(-a)m = -(am)$. By (ii) and axiom (i): $0 = a0 = a(m + (-m)) = am + a(-m)$, so $a(-m) = -(am)$. □

**Definition 4** (4.8 — Submodule)**.** *A subset $N$ of an $R$-module $M$ is an* $R$-submodule *of $M$ if:*

*(i)* $N \neq \emptyset$*;*

*(ii)* $a - b \in N$ *for all* $a, b \in N$*;*

*(iii)* $ra \in N$ *for all* $a \in N$*,* $r \in R$*.*

**Theorem 17** (4.12)**.** *Let $(N_i)_{i \in I}$ be a family of $R$-submodules of an $R$-module $M$. Then $\bigcap_{i \in I} N_i$ is also an $R$-submodule of $M$.*

*Proof.* Since $0 \in N_i$ for all $i$, we have $0 \in \bigcap N_i$, so it is non-empty. Let $x, y \in \bigcap N_i$ and $r \in R$. For each $i$, since $N_i$ is a submodule, $x - y \in N_i$ and $rx \in N_i$. Hence $x - y$ and $rx$ lie in $\bigcap N_i$. □

**Definition 5** (4.13 — Generated submodule, cyclic module)**.** *Let $S \subseteq M$. The family $\mathcal{A} = \{N \mid N \text{ is an } R\text{-submodule of } M \text{ containing } S\}$ is non-empty (since $M \in \mathcal{A}$), so by Theorem 4.12 the intersection $\bigcap_{N \in \mathcal{A}} N$ is an $R$-submodule. It is the* smallest *submodule of $M$ containing $S$, called the submodule* generated by *$S$ and denoted $(S)$. If $S = \{x_1, \ldots, x_n\}$ we write $(x_1, \ldots, x_n)$. An $R$-module $M$ is* cyclic *if $M = (x)$ for some $x \in M$.*

**Theorem 18** (4.15)**.** *Let $R$ be a ring with unity and $x_1, \ldots, x_n \in M$. Then*

$$(x_1, \ldots, x_n) = \sum_{i=1}^{n} Rx_i := \{r_1 x_1 + \cdots + r_n x_n \mid r_i \in R\}.$$

*Proof.* ($\supseteq$) The submodule $(x_1, \ldots, x_n)$ contains each $x_i$. By closure under scalar multiplication it contains $rx_i$ for all $r \in R$, and by closure under addition it contains every sum $r_1 x_1 + \cdots + r_n x_n$. Hence $\sum Rx_i \subseteq (x_1, \ldots, x_n)$.

($\subseteq$) We show $\sum Rx_i$ is a submodule containing each $x_i$; then $(x_1, \ldots, x_n) \subseteq \sum Rx_i$ follows from minimality of $(x_1, \ldots, x_n)$.

Each $x_i \in \sum Rx_i$ since $x_i = 1 \cdot x_i$ (using unity). For the submodule conditions: $0 = \sum 0 \cdot x_i \in \sum Rx_i$, so it is non-empty. For $\sum r_i x_i, \sum s_i x_i \in \sum Rx_i$ and $r \in R$,

$$\sum r_i x_i - \sum s_i x_i = \sum (r_i - s_i) x_i \in \sum Rx_i, \qquad r \sum r_i x_i = \sum (r r_i) x_i \in \sum Rx_i. \quad \square$$

**Definition 6** (4.18 — Sum of submodules)**.** *Let $N_1, \ldots, N_k$ be $R$-submodules of $M$. Their* sum *$\sum_{i=1}^{k} N_i$ is the submodule of $M$ generated by $\bigcup_{i=1}^{k} N_i$, i.e. the smallest submodule of $M$ containing each $N_i$.*

# Lecture 5 — Modules: continuation

**Theorem 19** (4.19)**.** *Let $N_1, \ldots, N_k$ be $R$-submodules of an $R$-module $M$. Then*

$$\sum_{i=1}^{k} N_i = \{x_1 + \cdots + x_k \mid x_i \in N_i\}.$$

*Proof.* Let

$$A = \{x_1 + \cdots + x_k \mid x_i \in N_i\}.$$

It is straightforward to check that $A$ is a submodule of $M$, and it contains every $N_i$. Hence, by minimality,

$$\sum_{i=1}^{k} N_i \subseteq A.$$

Conversely, $\sum_{i=1}^{k} N_i$ contains every $N_i$ and is closed under addition, so it contains every element of $A$. Therefore

$$A \subseteq \sum_{i=1}^{k} N_i.$$

□

**Definition 7** (4.21 — Direct sum)**.** *The sum $N_1 + \cdots + N_k$ is* direct *if every element has a unique representation*

$$x = x_1 + \cdots + x_k, \qquad x_i \in N_i.$$

*In this case we write*

$$N_1 \oplus \cdots \oplus N_k.$$

**Theorem 20** (4.22)**.** *Let $N_1, \ldots, N_k$ be submodules of an $R$-module $M$. The following are equivalent:*

*(i)* $\sum_{i=1}^{k} N_i$ *is direct;*

*(ii) if* $\sum_{i=1}^{k} x_i = 0$*, where* $x_i \in N_i$*, then every* $x_i = 0$*;*

*(iii) for every $i$,*

$$N_i \cap \sum_{j \neq i} N_j = \{0\}.$$

*Proof.* (i) ⇒ (ii): Zero already has the representation $0 + \cdots + 0$, so uniqueness implies that every $x_i = 0$.

(ii) ⇒ (iii): If

$$x \in N_i \cap \sum_{j \neq i} N_j,$$

then $x = \sum_{j \neq i} x_j$ for some $x_j \in N_j$. Hence

$$x - \sum_{j \neq i} x_j = 0,$$

so (ii) gives $x = 0$.

(iii) ⇒ (i): Suppose

$$\sum_{i=1}^{k} x_i = \sum_{i=1}^{k} y_i, \qquad x_i, y_i \in N_i.$$

For every $i$,

$$x_i - y_i = -\sum_{j \neq i} (x_j - y_j) \in N_i \cap \sum_{j \neq i} N_j = \{0\}.$$

Thus $x_i = y_i$ for every $i$, so the representation is unique.

□

**Definition 8** (4.23 — $R$-homomorphism)**.** *Let $M$ and $N$ be $R$-modules. A map $f\colon M \to N$ is an* $R$-homomorphism *if, for all $x, y \in M$ and $r \in R$,*

$$f(x+y) = f(x) + f(y), \qquad f(rx) = rf(x).$$

*The set of all $R$-homomorphisms $M \to N$ is denoted*

$$\mathrm{Hom}_R(M, N).$$

*An $R$-homomorphism $M \to M$ is called an* endomorphism*, and*

$$\mathrm{End}_R(M) := \mathrm{Hom}_R(M, M).$$

**Theorem 21** (4.30)**.** *Let $M = \bigoplus_{i=1}^{k} M_i$ be a direct sum of $R$-modules. Then*

$$\mathrm{Hom}_R(M, M) \cong \big(\mathrm{Hom}_R(M_j, M_i)\big)_{1 \le i,j \le k}$$

*as rings, where the right-hand side is a ring $T$ of $k \times k$ matrices under the usual matrix addition and multiplication, with $(i, j)$ entry in $\mathrm{Hom}_R(M_j, M_i)$.*

*Proof.* Let $\lambda_j : M_j \to M$ be the natural inclusion and $\pi_i : M \to M_i$ the natural projection. Two key identities hold:

$$\sum_{\ell=1}^{k} \lambda_\ell \pi_\ell = \mathrm{id}_M, \qquad \pi_p \lambda_q = \delta_{pq}\, \mathrm{id}_{M_p}.$$

The first says every $x \in M$ equals the sum of its components; the second says projecting the $q$-th inclusion gives zero unless $p = q$.

Define $\sigma : \mathrm{Hom}_R(M, M) \to T$ by $\sigma(\phi) = (\pi_i \phi \lambda_j)_{ij}$.

*Ring homomorphism.* Additivity is immediate. For multiplication, using the first identity:

$$(\sigma(\phi)\sigma(\psi))_{ij} = \sum_{\ell} \pi_i \phi \lambda_\ell \pi_\ell \psi \lambda_j = \pi_i \phi \left( \sum_{\ell} \lambda_\ell \pi_\ell \right) \psi \lambda_j = \pi_i \phi\, \mathrm{id}_M\, \psi \lambda_j = (\sigma(\phi\psi))_{ij}.$$

*Injective.* Suppose $\sigma(\phi) = 0$, so $\pi_i \phi \lambda_j = 0$ for all $i, j$. Applying $\lambda_i$ on the left and summing over $i$ using the first identity:

$$\phi \lambda_j = \mathrm{id}_M \cdot \phi \lambda_j = \Big( \sum_i \lambda_i \pi_i \Big) \phi \lambda_j = \sum_i \lambda_i (\pi_i \phi \lambda_j) = 0.$$

Now applying $\pi_j$ on the right and summing over $j$:

$$\phi = \phi \cdot \mathrm{id}_M = \phi \Big( \sum_j \lambda_j \pi_j \Big) = \sum_j (\phi \lambda_j) \pi_j = 0.$$

*Surjective.* Given $f = (f_{ij}) \in T$, set $\phi = \sum_{i,j} \lambda_i f_{ij} \pi_j$. Using the second identity:

$$(\sigma(\phi))_{st} = \pi_s \phi \lambda_t = \sum_{i,j} \pi_s \lambda_i\, f_{ij}\, \pi_j \lambda_t = \sum_{i,j} \delta_{si}\, f_{ij}\, \delta_{jt} = f_{st}.$$

Hence $\sigma(\phi) = f$, so $\sigma$ is surjective. □

**Definition 9** (Quotient module)**.** *Let $N$ be an $R$-submodule of an $R$-module $M$. The set of cosets $M/N = \{a + N \mid a \in M\}$ becomes an $R$-module under*

$$(a + N) + (b + N) = (a + b) + N, \qquad r(a + N) = ra + N.$$

*It is called the* quotient module *of $M$ by $N$.*

**Theorem 22** (4.31 — Correspondence Theorem for Modules)**.** *Let $N$ be an $R$-submodule of an $R$-module $M$. Every $R$-submodule of $M/N$ is of the form $U/N$, where $U$ is an $R$-submodule of $M$ containing $N$.*

*Proof.* The canonical projection $f : M \to M/N$, $f(x) = x + N$, is an $R$-homomorphism:

$$f(x+y) = (x+y)+N = (x+N)+(y+N) = f(x)+f(y), \qquad f(rx) = rx+N = r(x+N) = rf(x).$$

Let $X$ be an $R$-submodule of $M/N$. The structure of the proof is: define $U$ as the preimage of $X$ under $f$, show $U$ is a submodule of $M$ containing $N$, and conclude $X = U/N$.

Set $U = f^{-1}(X) = \{x \in M \mid f(x) \in X\}$.

$N \subseteq U$. For any $x \in N$, $f(x) = x+N = 0+N = 0_{M/N} \in X$ (since $X$ is a submodule, it contains 0). Hence $x \in U$.

*$U$ is an $R$-submodule.* Since $0 \in N \subseteq U$, $U$ is non-empty. For $x, y \in U$ and $r \in R$, since $f$ is an $R$-homomorphism:

$$f(x - y) = f(x) - f(y) \in X, \qquad f(rx) = rf(x) \in X,$$

so $x - y \in U$ and $rx \in U$.

$X = U/N$. By definition of $U$, $f(U) \subseteq X$. For the reverse inclusion, take any $x \in X$; since $f$ is surjective there exists $y \in M$ with $f(y) = x$, so $y \in U$ and $x \in f(U)$. Hence $X = f(U) = U/N$. □

**Theorem 23** (4.32 — First Isomorphism Theorem for Modules)**.** *Let $f : M \to N$ be an $R$-homomorphism. Then $M/\ker f \cong f(M)$.*

*Proof.* Define $g : M/\ker f \to f(M)$ by $g(x + \ker f) = f(x)$. We know from group theory that $g$ is an isomorphism of abelian groups (well-defined and injective because $x + \ker f = y + \ker f \iff x - y \in \ker f \iff f(x) = f(y)$; surjective onto $f(M)$ by definition). It remains to check that $g$ respects the $R$-action:

$$g(r(x + \ker f)) = g(rx + \ker f) = f(rx) = rf(x) = rg(x + \ker f).$$ □

**Theorem 24** (Second Isomorphism Theorem for Modules)**.** *Let $U$ and $V$ be $R$-submodules of an $R$-module $M$. Then $(U + V)/V \cong U/(U \cap V)$.*

*Proof.* Define $f : U \to (U + V)/V$ by $f(u) = u + V$.

*$f$ is an $R$-homomorphism.* For $u_1, u_2 \in U$ and $r \in R$:

$$f(u_1 + u_2) = (u_1 + u_2) + V = (u_1 + V) + (u_2 + V) = f(u_1) + f(u_2),$$

$$f(ru) = ru + V = r(u + V) = rf(u).$$

*$f$ is surjective.* Any element of $(U + V)/V$ has the form $(u + v) + V = u + V = f(u)$ for some $u \in U$, $v \in V$ (since $v \in V$ means $v + V = V$).

$\ker f = U \cap V$. We have $f(u) = 0$ iff $u + V = V$ iff $u \in V$, so $\ker f = \{u \in U \mid u \in V\} = U \cap V$.

By the First Isomorphism Theorem, $U/(U \cap V) \cong (U + V)/V$. □

**Theorem 25** (Third Isomorphism Theorem for Modules)**.** *Let $U \subseteq V$ be $R$-submodules of an $R$-module $M$. Then $(M/U)/(V/U) \cong M/V$.*

*Proof.* Define $f : M/U \to M/V$ by $f(x+U) = x+V$.

*$f$ is well-defined.* If $x+U = y+U$ then $x-y \in U \subseteq V$, so $x+V = y+V$.

*$f$ is an $R$-homomorphism.* For $x+U, y+U \in M/U$ and $r \in R$:

$$f((x+U)+(y+U)) = f((x+y)+U) = (x+y)+V = (x+V)+(y+V) = f(x+U)+f(y+U),$$

$$f(r(x+U)) = f(rx+U) = rx+V = r(x+V) = rf(x+U).$$

*$f$ is surjective.* For any $x+V \in M/V$, $f(x+U) = x+V$.

$\ker f = V/U$. We have $f(x+U) = 0$ iff $x+V = V$ iff $x \in V$, so $\ker f = \{x+U \mid x \in V\} = V/U$.

By the First Isomorphism Theorem, $(M/U)/(V/U) \cong M/V$. □

**Theorem 26** (4.33)**.** *Let $M_1, \ldots, M_n$ be $R$-modules and $A_i \leq M_i$ an $R$-submodule for each $i$. Then*

$$\frac{M_1 \times \cdots \times M_n}{A_1 \times \cdots \times A_n} \cong \frac{M_1}{A_1} \times \cdots \times \frac{M_n}{A_n}.$$

*Since the product is finite, $\times$ and $\oplus$ agree here, so equivalently $(M_1 \oplus \cdots \oplus M_n)/(A_1 \oplus \cdots \oplus A_n) \cong M_1/A_1 \oplus \cdots \oplus M_n/A_n$.*

*Proof.* Define $f : M_1 \times \cdots \times M_n \to M_1/A_1 \times \cdots \times M_n/A_n$ by

$$f(m_1, \ldots, m_n) = (m_1 + A_1, \ldots, m_n + A_n).$$

*$f$ is an $R$-homomorphism.* Each coordinate is the quotient map $\pi_i : M_i \to M_i/A_i$, and operations in a product are coordinatewise, so $f$ respects addition and scalar multiplication because each $\pi_i$ does.

*$f$ is surjective.* Any element of the target is $(m_1+A_1, \ldots, m_n+A_n)$ for some $m_i \in M_i$, and this equals $f(m_1, \ldots, m_n)$.

$\ker f = A_1 \times \cdots \times A_n$. We have $f(m_1, \ldots, m_n) = 0$ iff $m_i + A_i = A_i$ for every $i$, iff $m_i \in A_i$ for every $i$.

By the First Isomorphism Theorem, $(M_1 \times \cdots \times M_n)/(A_1 \times \cdots \times A_n) \cong M_1/A_1 \times \cdots \times M_n/A_n$. □

# Lecture 6 — Simple, Completely Reducible, and Free Modules

**Definition 10** (5.1 — Simple module)**.** *Let $R$ be a ring with unity. An $R$-module $M$ is called* simple *(or* irreducible*) if $M \neq 0$ and the only $R$-submodules of $M$ are $0$ and $M$.*

**Theorem 27** (5.5 — Characterisation of Simple Modules)**.** *Let $R$ be a ring with unity and $M$ an $R$-module. The following are equivalent:*

*(i) $M$ is simple;*

*(ii) $M \neq 0$, and $M$ is generated by any $0 \neq x \in M$;*

*(iii) $M \cong R/I$, where $I$ is a maximal left ideal of $R$.*

*Proof.* (i) $\Rightarrow$ (ii): Let $0 \neq x \in M$. Then $Rx$ is a non-zero submodule of $M$, so $Rx = M$ by simplicity.

(ii) $\Rightarrow$ (iii): Take $0 \neq x \in M$ and define $f : R \to M$ by $f(r) = rx$. This is a surjective $R$-homomorphism, so letting $I = \ker f$, the First Isomorphism Theorem gives $R/I \cong M$. Note that (ii) implies $M$ is simple: if $0 \neq N \leq M$ and $0 \neq x \in N$, then $M = Rx \subseteq N$. Hence $R/I \cong M$ is simple. By the Correspondence Theorem (4.31), any left ideal $J$ with $I \subseteq J \subseteq R$ corresponds to a submodule $J/I$ of $R/I$. Simplicity forces $J/I = 0$ or $J/I = R/I$, i.e. $J = I$ or $J = R$, so $I$ is maximal.

(iii) $\Rightarrow$ (i): Let $0 \neq N \leq M \cong R/I$. By the Correspondence Theorem (4.31), $N \cong J/I$ for some left ideal $J$ with $I \subsetneq J \subseteq R$. Since $I$ is maximal, $J = R$, so $N = M$. □

**Theorem 28** (5.6 — Schur's Lemma)**.** *Let $M$ be a simple $R$-module. Then* $\mathrm{Hom}_R(M, M)$ *is a division ring.*

*Proof.* Let $0 \neq f \in \mathrm{Hom}_R(M, M)$. We show $f$ is invertible and $f^{-1} \in \mathrm{Hom}_R(M, M)$.

*$f$ is bijective.* Since $\ker f$ and $\mathrm{im}\, f$ are submodules of $M$, simplicity forces each to be 0 or $M$. As $f \neq 0$, we have $\ker f \neq M$ and $\mathrm{im}\, f \neq 0$, so $\ker f = 0$ (injective) and $\mathrm{im}\, f = M$ (surjective).

*$f^{-1}$ is an $R$-homomorphism.* We cannot push $f^{-1}$ through operations directly, but we can push $f$ through them. The strategy is: to prove $f^{-1}(u) = v$, it suffices to show $f(u) = f(v)$, since $f$ is injective. For $x, y \in M$ and $r \in R$:

$$f\big(f^{-1}(x+y)\big) = x + y = f(f^{-1}(x)) + f(f^{-1}(y)) = f\big(f^{-1}(x) + f^{-1}(y)\big),$$
$$f\big(f^{-1}(rx)\big) = rx = rf(f^{-1}(x)) = f\big(rf^{-1}(x)\big),$$

where we used that $f$ is itself an $R$-homomorphism in the last step of each line. Injectivity of $f$ gives $f^{-1}(x+y) = f^{-1}(x) + f^{-1}(y)$ and $f^{-1}(rx) = rf^{-1}(x)$. □

**Corollary 5** (Non-isomorphic simple modules)**.** *Let $M$ and $N$ be simple $R$-modules. If $M \ncong N$, then*

$$\mathrm{Hom}_R(M, N) = 0.$$

*Proof.* Let $f : M \to N$ be a nonzero $R$-homomorphism. Since $\ker f$ is a submodule of the simple module $M$ and $f \neq 0$, we have $\ker f = 0$. Since $\mathrm{im}\, f$ is a nonzero submodule of the simple module $N$, we have $\mathrm{im}\, f = N$. Thus $f$ is an isomorphism, contradicting $M \ncong N$. Therefore every homomorphism $M \to N$ is zero, so $\mathrm{Hom}_R(M, N) = 0$. □

**Definition 11** (5.7 — Completely reducible module)**.** *Let $R$ be a ring with unity. An $R$-module $M$ is called* completely reducible *(or* semisimple*) if $M = \sum_{\alpha \in \Lambda} M_\alpha$, where each $M_\alpha$ is a simple $R$-submodule.*

**Theorem 29** (5.9)**.** *Let $M = \sum_{\alpha \in \Lambda} M_\alpha$ be a sum of simple $R$-submodules, and let $K$ be a submodule of $M$. Then there exists $\Lambda' \subseteq \Lambda$ such that $M = K \oplus \big(\bigoplus_{\alpha \in \Lambda'} M_\alpha\big)$.*

*Proof.* Let $S$ be the family of subsets $A \subseteq \Lambda$ satisfying two conditions: $\bigoplus_{\alpha \in A} M_\alpha$ is a direct sum, and $K \cap \bigoplus_{\alpha \in A} M_\alpha = 0$. Together these mean $K \oplus \bigoplus_{\alpha \in A} M_\alpha$ is a genuine direct sum. Since $\emptyset$ satisfies both conditions trivially, $S \neq \emptyset$. Partially order $S$ by inclusion; every chain $(A_i)$ in $S$ has upper bound $\bigcup A_i$ in $S$. By Zorn's Lemma, $S$ has a maximal element $A$.

Set $N = K \oplus \bigoplus_{\alpha \in A} M_\alpha$. We claim $N = M$. Fix any $\beta \in \Lambda$. Since $M_\beta$ is simple, $M_\beta \cap N$ is either 0 or $M_\beta$. If $M_\beta \cap N = 0$, then $M_\beta$ intersects neither $K$ nor $\bigoplus_{\alpha \in A} M_\alpha$, so $A \cup \{\beta\}$ would satisfy both conditions in $S$, contradicting maximality of $A$. Hence $M_\beta \cap N = M_\beta$, i.e. $M_\beta \subseteq N$, for every $\beta \in \Lambda$. Since $M = \sum_\beta M_\beta$, we conclude $N = M$. □

**Corollary 6** (5.10)**.** *If $M = \sum_{\alpha\in\Lambda} M_\alpha$ is a sum of simple $R$-submodules, then there exists $\Lambda' \subseteq \Lambda$ such that $M = \bigoplus_{\alpha\in\Lambda'} M_\alpha$.*

*Proof.* Apply Theorem 5.9 with $K = \{0\}$. □

**Definition 12** (5.11 — Linear independence and basis)**.** *A sequence $x_1, \ldots, x_n$ in an $R$-module $M$ is* linearly independent *if $\sum_{i=1}^n a_i x_i = 0$ implies $a_1 = \cdots = a_n = 0$ for any $a_i \in R$. A subset $S \subseteq M$ is linearly independent if every finite sequence of distinct elements of $S$ is linearly independent.*

*A subset $B \subseteq M$ is a* basis *of $M$ if $B$ generates $M$ and $B$ is linearly independent. An $R$-module that has a basis is called a* free module*.*

# Lecture 7 — Free Modules and Chain Conditions

**Theorem 30** (5.19)**.** *Let $M$ be a free $R$-module with basis $\{f_1, \ldots, f_n\}$. Then $M \cong R^n$ as $R$-modules.*

*Proof.* Define $\varphi : M \to R^n$ by $\varphi(\sum_{i=1}^n r_i f_i) = (r_1, \ldots, r_n)$, i.e. read off the coordinates in the basis. Since every element of $M$ has a *unique* representation as $\sum r_i f_i$ (by linear independence of the $f_i$), $\varphi$ is well-defined. It is an $R$-homomorphism: $\varphi(r \cdot \sum s_i f_i) = \varphi(\sum (rs_i) f_i) = (rs_1, \ldots, rs_n) = r \cdot (s_1, \ldots, s_n) = r \cdot \varphi(\sum s_i f_i)$, where the last scalar multiplication is coordinate-wise in $R^n$. It is injective (if $(r_1, \ldots, r_n) = 0$ then each $r_i = 0$, so $m = 0$), and surjective (every $(r_1, \ldots, r_n) \in R^n$ is the image of $\sum r_i f_i$). □

**Theorem 31** (5.20)**.** *Let $M$ be a finitely generated free $R$-module. Then every basis of $M$ is finite.*

*Proof.* Let $\{e_i\}_{i\in\Lambda}$ be a basis and $\{x_1, \ldots, x_n\}$ a finite generating set. Since elements of a module are finite sums by definition, each $x_j$ involves only finitely many nonzero coefficients, so is supported on a finite subset $S_j \subseteq \Lambda$. Every basis element must appear in the expression of some generator (otherwise it could not be reached), so $\Lambda \subseteq S_1 \cup \cdots \cup S_n$, a finite union of finite sets. Hence $\Lambda$ is finite. □

**Theorem 32** (5.21)**.** *Let $M$ be a finitely generated free module over a commutative ring $R$. Then all bases of $M$ have the same number of elements.*

*Proof.* Suppose $M$ has two bases of sizes $m$ and $n$ respectively. By Theorem 5.19, $M \cong R^m$ and $M \cong R^n$, so $R^m \cong R^n$. It suffices to show $m = n$.

Assume for contradiction $m < n$. Let $\phi : R^m \to R^n$ be an $R$-isomorphism with inverse $\psi = \phi^{-1}$. Let $(e_1, \ldots, e_m)$ and $(f_1, \ldots, f_n)$ be the standard bases of $R^m$ and $R^n$. Since $\phi$ is an $R$-homomorphism, it is determined by where it sends each $e_i$, and each image lands in $R^n$, so we can write

$$\phi(e_i) = \sum_{j=1}^{n} a_{ji} f_j, \qquad \psi(f_j) = \sum_{k=1}^{m} b_{kj} e_k.$$

Let $A = (a_{ji})$ be the resulting $n \times m$ matrix and $B = (b_{kj})$ the $m \times n$ matrix. Since $\psi\phi = \mathrm{id}_{R^m}$ and $\phi\psi = \mathrm{id}_{R^n}$, substituting the expressions above and using linear independence of the bases gives $BA = I_m$ and $AB = I_n$.

Pad these to $n \times n$ matrices by setting $A' = \begin{pmatrix} A & 0 \end{pmatrix}$ and $B' = \begin{pmatrix} B \\ 0 \end{pmatrix}$. Then $A'B' = I_n$, so $\det(A'B') = 1$. But $B'A' = \begin{pmatrix} I_m & 0 \\ 0 & 0 \end{pmatrix}$, so $\det(B'A') = 0$. Since $R$ is commutative, determinants multiply, giving

$$1 = \det(A'B') = \det(A')\det(B') = \det(B'A') = 0,$$

a contradiction. By symmetry $n < m$ also fails, so $m = n$. □

**Definition 13** (5.22 — Rank)**.** *The number of elements in any basis of a finitely generated free module $M$ over a commutative ring $R$ is called the* rank *of $M$, written* $\mathrm{rk}(M)$*. If $R$ is a field, the rank is called the* dimension*.*

**Lemma 2** (5.24)**.** *Every finitely generated $R$-module is a homomorphic image of a finitely generated free module.*

*Proof.* Let $M$ be generated by $x_1, \ldots, x_n$. Define $\phi : R^n \to M$ by $\phi(r_1, \ldots, r_n) = \sum_{i=1}^{n} r_i x_i$. This is a surjective $R$-homomorphism (surjective since the $x_i$ generate $M$). The generators $x_i$ may satisfy relations, so $\phi$ need not be injective — but by the First Isomorphism Theorem $M \cong R^n / \ker \phi$, expressing $M$ as a quotient of the free module $R^n$. □

**Lemma 3** (5.25)**.** *Let $R$ be a PID and $F$ a free $R$-module of rank $n$. Then every submodule $K \leq F$ is free of rank $m \leq n$.*

*Proof.* By Theorem 5.19, $F \cong R^n$, so we work with $K \leq R^n$ and induct on $n$. The base case $n = 0$ is trivial since $K = 0$ is free on the empty set.

Define $\pi : R^n \to R$ by $\pi(x_1, \ldots, x_n) = x_1$, and restrict it to $K$.

*Case 1:* $\pi|_K = 0$. Every element of $K$ has first coordinate zero, so $K \leq \{0\} \times R^{n-1} \cong R^{n-1}$. By induction $K$ is free of rank $\leq n - 1 \leq n$.

*Case 2:* $\pi|_K \neq 0$. The image $\pi(K)$ is a nonzero ideal of $R$. Since $R$ is a PID, $\pi(K) = Ra$ for some $a \neq 0$. Pick $k \in K$ with $\pi(k) = a$.

We claim $K = Rk \oplus \ker(\pi|_K)$, where $\ker(\pi|_K) = \{x \in K \mid x_1 = 0\}$ is the set of elements of $K$ with first coordinate zero, so $\ker(\pi|_K) \leq \{0\} \times R^{n-1} \cong R^{n-1}$.

$K = Rk + \ker(\pi|_K)$*:* For any $x \in K$, write $\pi(x) = ba$ for some $b \in R$. Then $\pi(x - bk) = ba - ba = 0$, so $x - bk \in \ker(\pi|_K)$, and $x = bk + (x - bk)$.

*The sum is direct:* If $ck \in \ker(\pi|_K)$, then $0 = \pi(ck) = ca$. Since $R$ is a PID (hence an integral domain) and $a \neq 0$, we get $c = 0$.

By induction $\ker(\pi|_K)$ is free of rank $s \leq n - 1$, and $Rk \cong R$ is free of rank 1. Hence $K \cong R^s \oplus R = R^{s+1}$, which has rank $m = s + 1 \leq n$. □

# Lecture 8 — Noetherian and Artinian Modules

**Definition 14** (6.1 — Noetherian and Artinian modules)**.** *An $R$-module $M$ is called* Noetherian *if every ascending chain of submodules*

$$M_1 \subseteq M_2 \subseteq M_3 \subseteq \cdots$$

*stabilises, i.e. there exists $k$ such that $M_k = M_{k+1} = \cdots$. This is called the* ascending chain condition *(ACC). An $R$-module $M$ is called* Artinian *if every descending chain of submodules stabilises (*descending chain condition*, DCC).*

**Theorem 33** (6.5)**.** *Let $M$ be an $R$-module. The following are equivalent:*

*(i) $M$ is Noetherian;*

*(ii) every submodule of $M$ is finitely generated;*

*(iii) every non-empty family $S$ of submodules of $M$ has a maximal element.*

*Proof.* (i) $\Rightarrow$ (ii): Let $N \leq M$ and suppose $N$ is not finitely generated. For any finite choice $a_1, \ldots, a_k \in N$ we have $N \neq (a_1, \ldots, a_k)$, so we can always pick $a_{k+1} \in N \setminus (a_1, \ldots, a_k)$. This produces an infinite strictly ascending chain $(a_1) \subsetneq (a_1, a_2) \subsetneq \cdots$, contradicting that $M$ is Noetherian.

(ii) $\Rightarrow$ (iii): Suppose $S$ has no maximal element. Then starting from any $N_0 \in S$ we can always find a strictly larger member, producing an infinite strictly ascending chain $N_0 \subsetneq N_1 \subsetneq N_2 \subsetneq \cdots$. Let $N = \bigcup_i N_i$. For $x, y \in N$ and $r \in R$, both $x$ and $y$ lie in some $N_\ell$ (take the larger index), so $x - y, rx \in N_\ell \subseteq N$; hence $N$ is a submodule. By (ii), $N = (a_1, \ldots, a_n)$ for some $a_j \in N$. Each $a_j$ lies in some $N_{u_j}$, so all generators lie in $N_\mu$ for $\mu = \max\{u_1, \ldots, u_n\}$. But then $N = (a_1, \ldots, a_n) \subseteq N_\mu \subseteq N$, so $N_\mu = N$, contradicting that the chain is strictly ascending.

(iii) $\Rightarrow$ (i): Given an ascending chain $M_1 \subseteq M_2 \subseteq \cdots$, by (iii) the family $\{M_i\}$ has a maximal element $M_k$. Then $M_k = M_{k+1} = \cdots$, so the chain stabilises. $\square$

**Definition 15** (6.6 — Finitely cogenerated)**.** *An $R$-module $M$ is* finitely cogenerated *if for every family $(M_\alpha)_{\alpha \in \Lambda}$ of submodules of $M$,*

$$\bigcap_{\alpha \in \Lambda} M_\alpha = 0 \implies \bigcap_{\alpha \in \Lambda'} M_\alpha = 0$$

*for some finite subset $\Lambda' \subseteq \Lambda$.*

**Theorem 34** (6.8)**.** *Let $M$ be an $R$-module. The following are equivalent:*

*(i) $M$ is Artinian;*

*(ii) every quotient module of $M$ is finitely cogenerated;*

*(iii) every non-empty family $S$ of submodules of $M$ has a minimal element.*

*Proof.* (i) $\Rightarrow$ (ii): Let $N \leq M$ and take any family $(M_\alpha/N)_{\alpha \in \Lambda}$ of submodules of $M/N$ with $\bigcap_{\alpha \in \Lambda} M_\alpha/N = 0$, i.e. $\bigcap_{\alpha \in \Lambda} M_\alpha = N$. Consider the family of finite intersections $\left\{\bigcap_{\alpha \in \Lambda'} M_\alpha \mid \Lambda' \subseteq \Lambda \text{ finite}\right\}$. As we add more submodules to the intersection it can only get smaller, so this is a descending family of submodules of $M$. Since $M$ is Artinian every descending chain stabilises, so there exists a finite $\Lambda' \subseteq \Lambda$ such that

$$\bigcap_{\alpha \in \Lambda'} M_\alpha = \bigcap_{\alpha \in \Lambda} M_\alpha = N.$$

Hence $\bigcap_{\alpha \in \Lambda'} M_\alpha/N = 0$, so $M/N$ is finitely cogenerated.

(ii) $\Rightarrow$ (iii): Suppose $S$ has no minimal element. Then starting from any $N_0 \in S$ we can always find a strictly smaller member, producing an infinite strictly descending chain $N_0 \supsetneq N_1 \supsetneq N_2 \supsetneq \cdots$. Let $N = \bigcap_i N_i$, which is a submodule of $M$. By (ii), $M/N$ is finitely cogenerated. The family $(N_i/N)$ of submodules of $M/N$ satisfies $\bigcap_i N_i/N = N/N = 0$, so by finite cogeneration there exists a finite subset, say $\{N_0, \ldots, N_k\}$, with $\bigcap_{i=0}^{k} N_i/N = 0$,

i.e. $\bigcap_{i=0}^{k} N_i = N$. Since the chain is descending, $\bigcap_{i=0}^{k} N_i = N_k$, so $N_k = N \subseteq N_{k+1} \subsetneq N_k$, a contradiction.

(iii) $\Rightarrow$ (i): Given a descending chain $M_1 \supseteq M_2 \supseteq \cdots$, by (iii) the family $\{M_i\}$ has a minimal element $M_k$. Then $M_k = M_{k+1} = \cdots$, so the chain stabilises. $\square$

**Theorem 35** (6.9)**.** *Let $M$ be a Noetherian (respectively Artinian) $R$-module and $K \leq M$ a submodule. Then $K$ and $M/K$ are both Noetherian (respectively Artinian).*

*Proof. Noetherian case.* By Theorem 6.5, $M$ is Noetherian if and only if every submodule of $M$ is finitely generated. Every submodule of $K$ is also a submodule of $M$, hence finitely generated. By Theorem 6.5 again, $K$ is Noetherian. For $M/K$: by Theorem 4.31 (Correspondence Theorem), every submodule of $M/K$ is of the form $L/K$ for some submodule $L$ with $K \subseteq L \subseteq M$. Since $L$ is a submodule of $M$, it is finitely generated, and hence $L/K$ is finitely generated. By Theorem 6.5, $M/K$ is Noetherian.

*Artinian case.* Any descending chain $K \supseteq L_1 \supseteq L_2 \supseteq \cdots$ of submodules of $K$ is also a descending chain in $M$, so it stabilises. Hence $K$ is Artinian. For $M/K$: a descending chain $M/K \supseteq L_1/K \supseteq L_2/K \supseteq \cdots$ of submodules of $M/K$ corresponds via Theorem 4.31 to a descending chain $M \supseteq L_1 \supseteq L_2 \supseteq \cdots$ in $M$, which stabilises. Hence $M/K$ is Artinian. $\square$

**Theorem 36** (6.10)**.** *Let $M$ be an $R$-module and $N \leq M$ a submodule. Then $M$ is Noetherian (respectively Artinian) if and only if both $N$ and $M/N$ are Noetherian (respectively Artinian).*

*Proof.* ($\Rightarrow$) This is Theorem 6.9.

($\Leftarrow$) *Noetherian case.* Assume $N$ and $M/N$ are Noetherian. Let $K \leq M$ be any submodule; we show $K$ is finitely generated, which by Theorem 6.5 gives that $M$ is Noetherian.

By the Second Isomorphism Theorem, $K/(N \cap K) \cong (K+N)/N$. Since $(K+N)/N$ is a submodule of $M/N$ and $M/N$ is Noetherian, it is finitely generated by Theorem 6.5, and hence so is $K/(N \cap K)$. Say $K/(N \cap K)$ is generated by $\bar{x}_1, \ldots, \bar{x}_m$ with $x_i \in K$. Then

$$K = (x_1, \ldots, x_m) + (N \cap K).$$

Since $N \cap K$ is a submodule of $N$ and $N$ is Noetherian, $N \cap K$ is finitely generated by Theorem 6.5, say by $y_1, \ldots, y_n \in N \cap K$. Therefore

$$K = (x_1, \ldots, x_m, y_1, \ldots, y_n),$$

so $K$ is finitely generated and $M$ is Noetherian.

*Artinian case.* Assume $N$ and $M/N$ are Artinian. Let $M \supseteq K_1 \supseteq K_2 \supseteq \cdots$ be a descending chain in $M$. Consider the two induced chains

$$K_1 \cap N \supseteq K_2 \cap N \supseteq \cdots \quad \text{in } N, \qquad \frac{K_1+N}{N} \supseteq \frac{K_2+N}{N} \supseteq \cdots \quad \text{in } M/N.$$

Both stabilise (since $N$ and $M/N$ are Artinian), say $K_i \cap N = K_p \cap N$ for all $i \geq p$, and $(K_i+N)/N = (K_s+N)/N$ (i.e. $K_i + N = K_s + N$) for all $i \geq s$. Set $q = \max\{p, s\}$. Then for all $i \geq q$:

$$K_i \cap N = K_q \cap N \quad \text{and} \quad K_i + N = K_q + N.$$

We claim $K_i = K_q$. Since the chain is descending and $i \geq q$, we already have $K_i \subseteq K_q$. For the reverse inclusion, take any $x \in K_q$. By the Second Isomorphism Theorem applied to $K_q$ and $N$:

$$K_q/(K_q \cap N) \cong (K_q + N)/N.$$

In concrete terms, since $K_q + N = K_i + N$, every element of $K_q$ can be written as an element of $K_i$ plus an element of $N$: there exist $y \in K_i$ and $n \in N$ such that $x = y + n$. Then

$$n = x - y \in K_q,$$

because $x \in K_q$ and $y \in K_i \subseteq K_q$. Hence $n \in K_q \cap N = K_i \cap N \subseteq K_i$. Therefore $x = y + n \in K_i$, giving $K_q \subseteq K_i$ and thus $K_i = K_q$. The original chain stabilises, so $M$ is Artinian. □

**Corollary 7** (6.11)**.** *If $M_1, \ldots, M_n$ are Noetherian (respectively Artinian) $R$-modules, then $M_1 \oplus \cdots \oplus M_n$ is Noetherian (respectively Artinian).*

*Proof.* By induction it suffices to treat the case of two modules $M$ and $N$. By Theorem 6.10, to show $M \oplus N$ is Noetherian (respectively Artinian) it suffices to find a single submodule of $M \oplus N$ such that both the submodule and its quotient module are Noetherian (respectively Artinian). Take $M \oplus 0 \leq M \oplus N$. Then:

- the submodule $M \oplus 0 \cong M$ is Noetherian (respectively Artinian) by assumption;
- the quotient $(M \oplus N)/(M \oplus 0) \cong N$ is Noetherian (respectively Artinian) by assumption.

Hence by Theorem 6.10, $M \oplus N$ is Noetherian (respectively Artinian). □

**Definition 16** (6.12 — Noetherian and Artinian rings)**.** *A ring $R$ is called* left Noetherian *(respectively* left Artinian*) if $R$, regarded as a left $R$-module, is Noetherian (respectively Artinian). By default, Noetherian (respectively Artinian) ring means left Noetherian (respectively left Artinian).*

**Theorem 37** (6.13)**.** *Let $R$ be a ring. The following are equivalent:*

*(i) $R$ is Noetherian (respectively Artinian);*

*(ii) every left ideal $A$ of $R$ is finitely generated (respectively every $R/A$ is finitely cogenerated);*

*(iii) every non-empty set of left ideals of $R$ has a maximal (respectively minimal) element.*

*Proof.* This is Theorem 6.5 (respectively 6.8) applied to $R$ regarded as a left $R$-module, where submodules of $R$ are precisely the left ideals of $R$. □

**Theorem 38** (6.19)**.** *Let $R_1, \ldots, R_n$ be Noetherian (respectively Artinian) rings. Then $R_1 \oplus \cdots \oplus R_n$ is Noetherian (respectively Artinian).*

*Proof.* This follows immediately from Corollary 6.11, viewing each $R_i$ as an $R$-module. □

**Product of ideals.** For ideals $A$ and $B$ in a ring $R$, define

$$AB = \left\{ \sum_{i=1}^{n} a_i b_i \;\middle|\; n \in \mathbb{N},\, a_i \in A,\, b_i \in B \right\}.$$

Then $AB$ is again an ideal of $R$.

**Definition 17** (6.20 — Prime ideal)**.** *Let $R$ be a ring. An ideal $P \neq R$ is* prime *if whenever $A$ and $B$ are ideals in $R$ with $AB \subseteq P$, then $A \subseteq P$ or $B \subseteq P$.*

(For commutative $R$ this recovers the classical definition: $ab \in P \Rightarrow a \in P$ or $b \in P$.)

**Theorem 39** (6.21)**.** *Let $R$ be a ring with unity. Then every maximal ideal is prime.*

*Proof.* Let $M$ be a maximal ideal and let $A, B$ be ideals with $AB \subseteq M$. Suppose $A \not\subseteq M$. Since $M$ is maximal, the only ideal strictly containing $M$ is $R$ itself, so $A + M = R$. Write $1 = a + m$ for some $a \in A$ and $m \in M$. For any $b \in B$,

$$b = 1 \cdot b = (a + m)b = ab + mb \in AB + MB \subseteq M + M = M.$$

Hence $B \subseteq M$, so $M$ is prime. □

*Note:* The converse is false: $(0)$ is a prime ideal in $\mathbb{Z}$ but not a maximal ideal.

# Lecture 9

**Lemma 4** (6.22)**.** *If $R$ is a Noetherian ring, then each ideal contains a finite product of prime ideals.*

*Proof.* Let $\mathcal{F}$ be the family of ideals of $R$ that do not contain any finite product of prime ideals. We want to show $\mathcal{F} = \emptyset$.

Suppose for a contradiction that $\mathcal{F} \neq \emptyset$. Since $R$ is Noetherian, by Theorem 6.13(iii) the family $\mathcal{F}$ has a maximal element, say $A$.

We claim $A$ is not a prime ideal. Indeed, any prime ideal $P$ contains the one-factor product $P$ itself, so no prime ideal can belong to $\mathcal{F}$.

Since $A$ is not prime, there exist ideals $B$ and $C$ with $BC \subseteq A$, but $B \not\subseteq A$ and $C \not\subseteq A$. Then $B + A \supsetneq A$ and $C + A \supsetneq A$, so by maximality of $A$ in $\mathcal{F}$, neither $B + A$ nor $C + A$ belongs to $\mathcal{F}$. Hence both contain finite products of prime ideals. But then $(B + A)(C + A)$ also contains a finite product of prime ideals, and

$$(B + A)(C + A) \subseteq BC + BA + AC + A^2 \subseteq A.$$

So $A$ contains a finite product of prime ideals, contradicting $A \in \mathcal{F}$. Hence $\mathcal{F} = \emptyset$. □

**Theorem 40** (6.23 — Hilbert's Basis Theorem)**.** *Let $R$ be a Noetherian ring. Then the polynomial ring $R[x]$ is also Noetherian.*

*Proof.* Suppose for a contradiction that $I \subseteq R[x]$ is a left ideal that is not finitely generated. We recursively construct a sequence of polynomials as follows. Set

$$J_0 = (0) \subseteq R[x].$$

Since $I$ is not finitely generated, $I \setminus J_0 \neq \emptyset$. Pick $f_0 \in I \setminus J_0$ of minimal degree (exists since degrees are non-negative integers). For $n \geq 1$, given $f_0, \ldots, f_{n-1}$, let

$$J_n = (f_0, \ldots, f_{n-1}) \subseteq R[x]$$

be the left ideal in $R[x]$ generated by $f_0, \ldots, f_{n-1}$. Since $I$ is not finitely generated, $I \setminus J_n \neq \emptyset$, so we may pick $f_n \in I \setminus J_n$ of minimal degree.

Since $J_n \subsetneq J_{n+1}$, the set $I \setminus J_{n+1} \subsetneq I \setminus J_n$, so $\deg(f_n)$ is non-decreasing in $n$.

For each $n$, let $a_n \in R$ denote the leading coefficient of $f_n$. Consider the ascending chain of left ideals in $R$:

$$L_1 \subseteq L_2 \subseteq \cdots, \qquad \text{where } L_n = (a_0, \ldots, a_{n-1}) \subseteq R.$$

Since $R$ is Noetherian, this chain stabilises: there exists $N \in \mathbb{N}$ such that $L_N = L_{N+1}$, that is,

$$a_N \in L_N = (a_0, \ldots, a_{N-1}) \subseteq R.$$

So we may write $a_N = \sum_{i<N} u_i a_i$ for some $u_i \in R$. Now define

$$g = \sum_{i<N} u_i x^{\deg(f_N)-\deg(f_i)} f_i \in J_N.$$

The exponents $\deg(f_N) - \deg(f_i) \geq 0$ since degrees are non-decreasing. The leading term of $u_i x^{\deg(f_N)-\deg(f_i)} f_i$ is $u_i a_i x^{\deg(f_N)}$, so the leading term of $g$ is

$$\left( \sum_{i<N} u_i a_i \right) x^{\deg(f_N)} = a_N x^{\deg(f_N)},$$

which equals the leading term of $f_N$. Hence $f_N - g$ has degree strictly less than $\deg(f_N)$.

Since $f_N \notin J_N$ and $g \in J_N$, we have $f_N - g \notin J_N$. Since $f_N, g \in I$, we have $f_N - g \in I$. Thus $f_N - g \in I \setminus J_N$.

- If $\deg(f_N) > 0$: then $f_N - g \in I \setminus J_N$ has degree strictly less than $\deg(f_N)$, contradicting the minimality of $\deg(f_N)$.
- If $\deg(f_N) = 0$: then all degrees are 0, so $f_N - g = 0$, meaning $f_N = g \in J_N$, contradicting $f_N \in I \setminus J_N$.

In both cases we reach a contradiction, so $I$ must be finitely generated and $R[x]$ is Noetherian. □

**Definition 18** (7.1 — Nilpotent elements and ideals)**.** *Let $R$ be a ring.*

- *An element $a \in R$ is* nilpotent *if $a^n = 0$ for some $n \in \mathbb{N}$.*
- *A left ideal $I$ of $R$ is a* nil ideal *if every element of $I$ is nilpotent.*
- *A left ideal $I$ of $R$ is* nilpotent *if $I^n = 0$ for some $n \in \mathbb{N}$, where $I^n$ denotes the set of all finite sums of products $i_1 i_2 \cdots i_n$ with each $i_k \in I$.*

*Every nilpotent ideal is nil (if $I^n = 0$ then any $a \in I$ satisfies $a^n \in I^n = 0$), but the converse fails in general.*

**Theorem 41** (7.5)**.** *If $J$ is a nil left ideal in an Artinian ring $R$, then $J$ is nilpotent.*

*Proof.* Suppose for a contradiction that $J^k \neq 0$ for all $k \in \mathbb{N}$. The family $(J, J^2, J^3, \ldots)$ is a descending chain of left ideals of $R$. Since $R$ is Artinian, this family has a minimal element, say $B = J^m$ for some $m$. Then

$$B^2 = J^{2m} \subseteq J^m = B,$$

and by minimality of $B$ we have $B^2 = B$.

Now consider the family

$$\mathcal{F} = \{A \mid A \text{ is a left ideal of } R,\ A \subseteq B,\ BA \neq 0\}.$$

Since $B^2 = B \neq 0$, we have $B \in \mathcal{F}$, so $\mathcal{F} \neq \emptyset$. Since $R$ is Artinian, $\mathcal{F}$ has a minimal element $A$.

Since $BA \neq 0$, there exists $a \in A$ with $Ba \neq 0$. Note that $Ba = \{ba \mid b \in B\}$ is a left ideal contained in $A$, and

$$B(Ba) = B^2 a = Ba \neq 0,$$

so $Ba \in \mathcal{F}$. By minimality of $A$ we conclude $Ba = A$.

Hence there exists $b \in B$ with $ba = a$. Applying $b$ repeatedly gives $b^i a = a$ for all $i \in \mathbb{N}$. But $b \in B \subseteq J$ and $J$ is nil, so $b^n = 0$ for some $n \in \mathbb{N}$. Then $a = b^n a = 0$, contradicting $Ba \neq 0$.

Therefore $J^k = 0$ for some $k \in \mathbb{N}$, i.e. $J$ is nilpotent. □

**Lemma 5** (7.6)**.** *Let $R$ be a Noetherian ring. Then the sum of nilpotent left ideals in $R$ is a nilpotent left ideal.*

*Proof.* Let $B = \sum_{i \in \Lambda} A_i$ be a sum of nilpotent left ideals. Since $R$ is Noetherian, $B$ is finitely generated as a left ideal, say $B = (x_1, \ldots, x_m)$. Each $x_j$ lies in a finite sum of the $A_i$'s, so $B$ is contained in a finite sum $A_1 + \cdots + A_n$ (reindexing if necessary), and hence $B = A_1 + \cdots + A_n$.

It remains to show $B$ is nilpotent. For each $i$, let $n_i \in \mathbb{N}$ be such that $A_i^{n_i} = 0$. Set $N = n_1 + \cdots + n_n$. Any product of $N$ elements from $B = A_1 + \cdots + A_n$, when expanded, is a sum of products of the form $a_{i_1} a_{i_2} \cdots a_{i_N}$ with each $a_{i_k} \in A_{i_k}$. By pigeonhole, among the $N$ indices $i_1, \ldots, i_N \in \{1, \ldots, n\}$, at least one value $j$ appears at least $n_j$ times. The consecutive factors from $A_j$ multiply to give an element of $A_j^{n_j} = 0$. Hence every such product is 0, so $B^N = 0$. □

# Lecture 10

**Theorem 42** (7.7)**.** *Let $R$ be a Noetherian ring having no non-zero nilpotent ideals. Then $R$ has no non-zero nil ideals.*

*(Proof non-examinable.)*

**Theorem 43** (7.8)**.** *Let $N$ be a nil ideal in a Noetherian ring $R$. Then $N$ is nilpotent.*

*Proof.* Let $T$ be the sum of all nilpotent ideals in $R$. By Lemma 7.6, $T$ is itself nilpotent, say $T^k = 0$.

We claim $R/T$ has no non-zero nilpotent ideals. Suppose $A/T$ is a nilpotent ideal in $R/T$, say $(A/T)^m = 0$. Then $(A^m + T)/T = 0$, so $A^m \subseteq T$. Since $T^k = 0$, we get $A^{mk} \subseteq T^k = 0$, so $A$ is nilpotent. Hence $A \subseteq T$, giving $A/T = 0$.

Since $R/T$ is Noetherian (a quotient of a Noetherian ring) and has no non-zero nilpotent ideals, Theorem 7.7 gives that $R/T$ has no non-zero nil ideals. Consider $(N+T)/T$: every element has the form $n + T$ for some $n \in N$ (since $t + T = T$ for $t \in T$), and since $N$ is nil there exists $k$ with $n^k = 0$, giving $(n + T)^k = n^k + T = T$. Hence $(N + T)/T$ is a nil ideal in $R/T$, so $(N + T)/T = 0$, which means $N \subseteq T$. Since $T$ is nilpotent, $N$ is nilpotent. □

**Lemma 6** (7.9)**.** *Let $A$ be a minimal left ideal in a ring $R$. Then either $A^2 = 0$ or $A = Re$ for some idempotent $e \in R$.*

*Proof.* Assume $A^2 \neq 0$.

*Step 1: Find $e \in A$ with $ea = a$.* Since $A^2 \neq 0$, there exists $a \in A$ with $Aa \neq 0$. The set $Aa = \{ca \mid c \in A\}$ is a left ideal contained in $A$ (since $A$ is a left ideal and $a \in A$). By minimality of $A$, we have $Aa = A$. In particular, there exists $e \in A$ with $ea = a$.

*Step 2: Show $e^2 = e$.* We compute $(e^2 - e)a = e(ea) - ea = ea - ea = 0$. To conclude $e^2 = e$, define the left annihilator of $a$ in $A$:

$$\mathrm{Ann}_A(a) := \{c \in A \mid ca = 0\}.$$

This is a left ideal of $R$ contained in $A$. Since $ea = a \neq 0$, we have $e \notin \mathrm{Ann}_A(a)$, so $\mathrm{Ann}_A(a) \neq A$. By minimality of $A$, we conclude $\mathrm{Ann}_A(a) = 0$. Since $e^2 - e \in A$ and $(e^2 - e)a = 0$, we get $e^2 - e \in \mathrm{Ann}_A(a) = 0$, hence $e^2 = e$.

*Step 3: Show $A = Re$.* Since $e \in A$ and $A$ is a left ideal, $Re \subseteq A$. Since $e \neq 0$ (as $ea = a \neq 0$), we have $Re \neq 0$. By minimality of $A$, $Re = A$. □

**Theorem 44** (7.10 — Wedderburn–Artin)**.** *Let $R$ be a left (or right) Artinian ring with unity and no non-zero nilpotent ideals. Then*

$$R \cong M_{n_1}(D_1) \oplus \cdots \oplus M_{n_k}(D_k)$$

*for some $n_1, \ldots, n_k \in \mathbb{N}$ and division rings $D_1, \ldots, D_k$.*

*Proof.* We build the result from two claims, then combine them.

**Claim 1.** *Each non-zero left ideal $A$ of $R$ is of the form $Re$ for some idempotent $e \in R$.*

By Lemma 7.9, every minimal left ideal is either nilpotent or of the form $Re$ for some idempotent $e$. Since $R$ has no non-zero nilpotent ideals, every minimal left ideal is of the form $Re$.

Now let $A$ be any non-zero left ideal. Since $R$ is Artinian, $A$ contains a minimal left ideal, hence a non-zero idempotent. Thus the family

$$\mathcal{F} = \{R(1 - f) \cap A \mid f \text{ a non-zero idempotent in } A\}$$

is non-empty, and by the Artinian condition has a minimal member $R(1 - e) \cap A$.

**Subclaim** *(non-examinable).* $R(1 - e) \cap A = 0$.

Invoking the subclaim: for any $a \in A$, $a(1 - e) \in R(1 - e) \cap A = 0$, so $a = ae \in Re$. Hence $A \subseteq Re \subseteq A$, giving $A = Re$.

**Claim 2.** *$R = Re_1 \oplus \cdots \oplus Re_n$ is a finite direct sum of minimal left ideals.*

Let $S$ be the sum of all minimal left ideals in $R$; since $R$ is Artinian it has at least one minimal left ideal, so $S \neq 0$. By Claim 1, $S = Re$ for some idempotent $e$.

Suppose $R(1-e) \neq 0$. Then $R(1-e)$ is a non-zero left ideal, so by the Artinian condition it contains a minimal left ideal $A$. But $A \subseteq S = Re$, giving $A \subseteq Re \cap R(1-e)$. For any $x \in Re \cap R(1-e)$, write $x = ae = b(1-e)$ for some $a, b \in R$. Then $xe = ae^2 = ae = x$, and also $xe = b(1-e)e = b(e-e^2) = 0$, so $x = 0$. Hence $Re \cap R(1-e) = 0$, giving $A = 0$, a contradiction.

Therefore $R(1-e) = 0$. Since $R$ has unity, $1 - e = 0$, i.e. $e = 1$, so $S = R$.

Thus $R$ is the sum of all its minimal left ideals. By Theorem 5.9 there exists a subfamily $(A_i)_{i \in \Lambda'}$ such that $R = \bigoplus_{i \in \Lambda'} A_i$. To see this is finite, write $1 = e_{i_1} + \cdots + e_{i_n}$ as a finite sum with $e_{i_j} \in A_{i_j}$. Then for any $r \in R$, $r = r \cdot 1 = re_{i_1} + \cdots + re_{i_n}$, so $R = Re_{i_1} \oplus \cdots \oplus Re_{i_n}$. After reindexing, $R = Re_1 \oplus \cdots \oplus Re_n$.

**Conclusion.** The plan: compute $\mathrm{End}_R(R) = \mathrm{Hom}_R(R, R)$ as a matrix ring using Theorem 4.30, use Schur's Lemma to see that it is block diagonal, and finally recover $R$ from $\mathrm{End}_R(R)$.

*Step 1: Group isomorphic summands.* By Claim 2, $R = Re_1 \oplus \cdots \oplus Re_n$ as left $R$-modules, where each $Re_i$ is a minimal left ideal, i.e. a simple left $R$-module. Sort the summands into isomorphism classes; say there are $k$ classes, of sizes $n_1, \ldots, n_k$, and choose a representative $N_j$ of the $j$-th class. Thus $N_i \ncong N_j$ for $i \neq j$, and every $Re_\ell$ is isomorphic to exactly one $N_j$. The order of summands in a direct sum is irrelevant, so we may reorder them so that each class is adjacent. Replacing each summand by its (isomorphic) representative then gives an isomorphism of left $R$-modules

$$R \;\cong\; M := \underbrace{(N_1 \oplus \cdots \oplus N_1)}_{n_1} \oplus \cdots \oplus \underbrace{(N_k \oplus \cdots \oplus N_k)}_{n_k}.$$

A module isomorphism $\theta : R \to M$ induces a ring isomorphism $\mathrm{End}_R(R) \to \mathrm{End}_R(M)$, $\varphi \mapsto \theta\varphi\theta^{-1}$. Hence $\mathrm{End}_R(R) \cong \mathrm{End}_R(M)$, and it suffices to compute $\mathrm{End}_R(M)$.

*Step 2: Apply Schur.* Since each $N_j$ is simple, Schur's Lemma 5.6 says every non-zero homomorphism between $N_i$ and $N_j$ is an isomorphism. Therefore

$$\mathrm{Hom}_R(N_i, N_j) = 0 \quad (i \neq j), \qquad D_j := \mathrm{End}_R(N_j) \text{ is a division ring.}$$

*Step 3: Compute* $\mathrm{End}_R(M)$. View $M$ as a direct sum of $n = n_1 + \cdots + n_k$ simple summands. By Theorem 4.30, $\mathrm{End}_R(M)$ is isomorphic to the ring of $n \times n$ matrices whose $(p, q)$ entry lies in $\mathrm{Hom}_R(q\text{-th summand}, p\text{-th summand})$. By Step 2, this entry is 0 if the two summands are copies of different $N_i, N_j$, and is an arbitrary element of $D_j$ if both are copies of $N_j$. Since each class is adjacent, these are exactly the block-diagonal matrices

$$\begin{pmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_k \end{pmatrix}, \qquad A_j \in M_{n_j}(D_j).$$

Block-diagonal matrices add and multiply block by block, $\mathrm{diag}(A_1, \ldots, A_k)\,\mathrm{diag}(B_1, \ldots, B_k) = \mathrm{diag}(A_1B_1, \ldots, A_kB_k)$, so sending a matrix to its tuple of blocks is a ring isomorphism. Hence

$$\mathrm{End}_R(R) \;\cong\; \mathrm{End}_R(M) \;\cong\; M_{n_1}(D_1) \oplus \cdots \oplus M_{n_k}(D_k).$$

*Step 4: Identify* $\mathrm{End}_R(R)$ *with* $R^{\mathrm{op}}$. Every $\varphi \in \mathrm{End}_R(R)$ is determined by $b := \varphi(1)$, since $\varphi(r) = \varphi(r \cdot 1) = r\varphi(1) = rb$. So every endomorphism is right multiplication $\rho_b : r \mapsto rb$ for a unique $b \in R$, and conversely each $\rho_b$ is an endomorphism. Composition reverses the order of multiplication: $(\rho_b \circ \rho_c)(r) = rcb$, so $\rho_b \circ \rho_c = \rho_{cb} = \rho_{b \cdot_{\mathrm{op}} c}$. Hence $b \mapsto \rho_b$ is a ring isomorphism $R^{\mathrm{op}} \cong \mathrm{End}_R(R)$, and combining with Step 3,

$$R^{\mathrm{op}} \;\cong\; M_{n_1}(D_1) \oplus \cdots \oplus M_{n_k}(D_k).$$

*Step 5: Pass to opposite rings.* Taking opposites of both sides (using $(R^{\mathrm{op}})^{\mathrm{op}} = R$), and noting that the opposite of a direct sum is the direct sum of the opposites,

$$R \;\cong\; M_{n_1}(D_1)^{\mathrm{op}} \oplus \cdots \oplus M_{n_k}(D_k)^{\mathrm{op}}.$$

Transposition gives $M_n(D)^{\mathrm{op}} \cong M_n(D^{\mathrm{op}})$, and $D^{\mathrm{op}}$ is again a division ring. Relabelling $D_j^{\mathrm{op}}$ as $D_j$, we conclude

$$R \;\cong\; M_{n_1}(D_1) \oplus \cdots \oplus M_{n_k}(D_k). \qquad \square$$

**Theorem 45** (7.11)**.** *Let $R$ be a left (or right) Artinian ring with unity and no non-zero nilpotent ideals. Then $R$ is also right and left Artinian, and right and left Noetherian.*

*(Non-examinable. Follows from Wedderburn–Artin and the fact that matrix rings over division rings are Noetherian and Artinian on both sides.)*

**Theorem 46** (7.13 — Maschke)**.** *If $G$ is a finite group, then*

$$\mathbb{C}[G] \cong M_{n_1}(\mathbb{C}) \oplus \cdots \oplus M_{n_k}(\mathbb{C})$$

*for some $n_1, \ldots, n_k \in \mathbb{N}$.*

*(Non-examinable. Follows from Wedderburn–Artin: one shows $\mathbb{C}[G]$ has no non-zero nilpotent ideals and is Artinian, then uses that the only division ring containing $\mathbb{C}$ in its centre as an algebraically closed field is $\mathbb{C}$ itself.)*

**Theorem 47** (7.14)**.** *Let $R$ be a ring with unity. The following are equivalent:*

*(i) $R$ is left Artinian with no non-zero nilpotent ideals;*

*(ii) $R$ is left Artinian with no non-zero nil ideals;*

*(iii) $R$ is a finite direct sum of minimal left ideals;*

*(iv) every left ideal of $R$ is of the form $Re$ for some idempotent $e$;*

*(v) $R$ is a finite direct sum of matrix rings over division rings.*

*(Non-examinable.)*

**Definition 19** (7.15 — Semisimple Artinian ring)**.** *A ring satisfying any of the equivalent conditions of Theorem 7.14 is called a* semisimple Artinian ring.

# Lecture 11 — Smith Normal Form over a PID

Throughout this chapter $R$ is a principal ideal domain with unity.

**Definition 20** (8.1 — Row module, column module)**.** *Let $A \in M_{m\times n}(R)$. The submodule of $R^n$ generated by the $m$ rows of $A$ is called the* row module *of $A$, denoted $\mathcal{R}(A)$. The submodule of $R^m$ generated by the $n$ columns of $A$ is called the* column module *of $A$, denoted $\mathcal{C}(A)$.*

*Since $\mathcal{R}(A)$ and $\mathcal{C}(A)$ are finitely generated submodules of the free modules $R^n$ and $R^m$ respectively, both are free by Lemma 5.25.*

**Definition 21** (8.2 — Row rank, column rank)**.** *Let $A \in M_{m\times n}(R)$. The rank of $\mathcal{R}(A)$ is called the* row rank *of $A$, and the rank of $\mathcal{C}(A)$ is called the* column rank *of $A$.*

**Theorem 48** (8.3)**.** *Let $A \in M_{m\times n}(R)$ and let $P$, $Q$ be invertible $m \times m$ and $n \times n$ matrices over $R$ respectively. Then row rank and column rank of $PAQ$ equal those of $A$.*

*(Proof non-examinable.)*

**Definition 22** (8.4/8.6 — Elementary operations and matrices)**.** *Let $A \in M_{m\times n}(R)$. There are three* elementary row operations*, each achieved by left-multiplying $A$ by an invertible* elementary matrix *(column operations are achieved by right-multiplying):*

*(i) Swap rows $i$ and $j$: multiply on the left by $S_{ij} = I - E_{ii} - E_{jj} + E_{ij} + E_{ji}$. Here $S_{ij}^{-1} = S_{ij}$.*

*(ii) Scale row $i$ by invertible $\alpha \in R$: multiply on the left by $M_i(\alpha) = I + (\alpha - 1)E_{ii}$. Here $M_i(\alpha)^{-1} = M_i(\alpha^{-1})$.*

*(iii) Add $\alpha \in R$ times row $j$ to row $i$: multiply on the left by $C_{ij}(\alpha) = I + \alpha E_{ij}$. Here $C_{ij}(\alpha)^{-1} = C_{ij}(-\alpha)$.*

*Since each elementary matrix is invertible, any sequence of elementary row and column operations transforms $A$ into an equivalent matrix $PAQ$.*

**Definition 23** (8.7 — Matrix equivalence)**.** *Two matrices $A, B \in M_{m\times n}(R)$ are* equivalent *if there exist invertible matrices $P \in M_m(R)$ and $Q \in M_n(R)$ such that $B = PAQ$.*

**Theorem 49** (8.8 — Smith Normal Form)**.** *Let $A \in M_{m\times n}(R)$ where $R$ is a PID. Then $A$ is equivalent to a matrix of the form*

$$\begin{pmatrix} a_1 & & & \\ & \ddots & & \\ & & a_r & \\ & & & 0 \end{pmatrix},$$

*where $a_1, \ldots, a_r \neq 0$ and $a_1 \mid a_2 \mid \cdots \mid a_r$.*

*Proof.* Define the *length* $\ell(a)$ of a non-zero $a \in R$ to be the number of prime factors in the factorisation $a = p_1 \cdots p_k$ counted with multiplicity, with the convention $\ell(u) = 0$ for units.

*Step 1: bring a minimal-length entry to position* $(1,1)$. If $A = 0$ the result is immediate. Otherwise, among all non-zero entries pick $a_{ij}$ with minimal $\ell(a_{ij})$. Row and column swaps bring it to position $(1,1)$.

*Step 2: clear the first row.* Fix any column $k \geq 2$. If $a_{11} \mid a_{1k}$, write $a_{1k} = a_{11}q$ and apply $C_k - qC_1$, zeroing the $(1,k)$ entry without changing $a_{11}$. If $a_{11} \nmid a_{1k}$, let $d = \gcd(a_{11}, a_{1k})$ and write $a_{11} = ds$, $a_{1k} = dt$. By Bézout there exist $u, v \in R$ with $us + vt = 1$, so the matrix $\left(\begin{smallmatrix} u & t \\ v & -s \end{smallmatrix}\right)$ is invertible (determinant $-1$). Right-multiplying $A$ by the $n \times n$ matrix with this $2 \times 2$ block in the $(1,k)$-positions and the identity elsewhere replaces the $(1,1)$ entry by $a_{11}u + a_{1k}v = d$ and the $(1,k)$ entry by $a_{11}t - a_{1k}s = 0$. Since $d$ properly divides $a_{11}$, we have $\ell(d) < \ell(a_{11})$. Repeating across all columns clears the first row.

*Step 3: clear the first column.* The identical argument applied to rows (left-multiplying by the analogous block matrices) clears all entries below $(1,1)$.

*Step 4: termination.* Clearing the first column may reintroduce non-zero entries in the first row, requiring another pass of Step 2. However this only occurs when $a_{11}$ does not divide some first-column entry, which triggers the Bézout case and strictly reduces $\ell(a_{11})$. Since $\ell(a_{11}) \geq 0$ is a non-negative integer, the alternating process terminates after at most $\ell(a_{11})$ restarts, yielding

$$P_1 A Q_1 = \begin{pmatrix} a_1 & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & A_1 & \\ 0 & & & \end{pmatrix},$$

where $A_1$ is an $(m-1) \times (n-1)$ submatrix.

*Step 5: induction.* Apply the same procedure to $A_1$ and continue. By induction on $m + n$ we obtain $PAQ = \operatorname{diag}(a_1, \ldots, a_r, 0, \ldots, 0)$.

*Step 6: enforce* $a_1 \mid a_2 \mid \cdots \mid a_r$. Suppose $a_1 \nmid a_2$. Add row 2 to row 1; the first row becomes $(a_1, a_2, 0, \ldots, 0)$. Applying Step 2 replaces $a_1$ by $\gcd(a_1, a_2)$, strictly reducing $\ell(a_1)$. Repeating forces $a_1 \mid a_2$, and similarly $a_1 \mid a_i$ for all $i$. Repeating the argument for $a_2$ in place of $a_1$, and so on, gives $a_i \mid a_{i+1}$ for all $i = 1, \ldots, r-1$. □

**Definition 24** (8.9 — Invariant factors)**.** *The non-zero diagonal entries $a_1, \ldots, a_r$ of the Smith normal form of $A$ are called the* invariant factors *of $A$. They are unique up to unit multipliers, and two $m \times n$ matrices are equivalent if and only if they have the same invariant factors.*

**Theorem 50** (8.10)**.** *Let $A$ be an $m \times n$ matrix over a PID. Then the row rank of $A$ equals the column rank of $A$.*

*Proof.* Choose invertible $P$, $Q$ such that $PAQ$ is in Smith normal form (Theorem 8.8). By Theorem 8.3, multiplying by invertible matrices preserves row and column rank, so

$$\text{row rank}(A) = \text{row rank}(PAQ) = r = \text{col rank}(PAQ) = \text{col rank}(A),$$

where $r$ is the number of non-zero diagonal entries. □

**Definition 25** (8.11 — Rank)**.** *The common value of the row rank and column rank of $A$ is called the* rank *of $A$, denoted* $\operatorname{rank}(A)$.

# Lecture 12 — Finitely Generated Modules over a PID

**Theorem 51** (9.1 — Decomposition Theorem)**.** *Let $R$ be a principal ideal domain, and let $M$ be a finitely generated $R$-module. Then*

$$M \cong R^s \oplus R/Ra_1 \oplus \cdots \oplus R/Ra_r,$$

*where $s, r \in \mathbb{N} \cup \{0\}$, the $a_i$ are non-zero non-units in $R$, and $a_i \mid a_{i+1}$ for $i = 1, \ldots, r-1$.*

*Proof.* Since $M$ is finitely generated, Lemma 5.24 gives a surjective $R$-homomorphism $\phi_0 : R^n \to M$. Let $K = \ker \phi_0$. By the First Isomorphism Theorem,

$$M \cong R^n/K.$$

Because $K$ is a submodule of the finite free module $R^n$ and $R$ is a PID, Lemma 5.25 says that $K$ is free of finite rank $m \leq n$. Choose an isomorphism $\phi : R^m \xrightarrow{\sim} K$. The composite

$$R^m \xrightarrow{\phi} K \hookrightarrow R^n$$

is represented by an $n \times m$ matrix $A$: its $j$th column is the vector $\phi(e_j) \in K \subseteq R^n$, where $e_j$ is the $j$th standard basis vector of $R^m$. Since $\phi$ maps onto $K$, the image of this matrix is exactly

$$AR^m = K.$$

By Smith Normal Form (Theorem 8.8), there are invertible matrices $P \in M_n(R)$ and $Q \in M_m(R)$ such that

$$PAQ = D = \operatorname{diag}(d_1, \ldots, d_k, 0, \ldots, 0), \qquad d_1 \mid \cdots \mid d_k,$$

where $d_1, \ldots, d_k$ are the nonzero diagonal entries. Since $Q$ is invertible, $QR^m = R^m$: every vector in $R^m$ is $Qv$ for some $v \in R^m$. Therefore

$$AQR^m = AR^m = K.$$

Multiplying this equality of submodules by $P$ gives

$$DR^m = PAQR^m = PK.$$

We now explicitly compare the quotients. Define

$$\overline{P} : R^n/K \longrightarrow R^n/PK, \qquad v + K \longmapsto Pv + PK.$$

This is well-defined: if $v + K = w + K$, then $v - w \in K$, so $P(v - w) \in PK$, and hence $Pv + PK = Pw + PK$. It is an isomorphism, with inverse $u + PK \mapsto P^{-1}u + K$. Thus, using $K = AR^m$ and $PK = PAQR^m = DR^m$,

$$M \cong R^n/K \cong R^n/PK = R^n/PAQR^m = R^n/DR^m.$$

To compute $DR^m$, write $v = (v_1, \ldots, v_m)^{\mathsf{T}}$. Then

$$Dv = (d_1v_1, \ldots, d_kv_k, 0, \ldots, 0)^{\mathsf{T}}.$$

The inputs $v_1, \ldots, v_k$ vary independently over $R$, so the first $k$ output coordinates vary independently over the ideals $Rd_1, \ldots, Rd_k$. Hence, as a submodule of $R^n = R \oplus \cdots \oplus R$,

$$DR^m = Rd_1 \oplus \cdots \oplus Rd_k \oplus \underbrace{0 \oplus \cdots \oplus 0}_{n-k}.$$

Applying Theorem 4.33 coordinate by coordinate gives

$$\frac{R^n}{DR^m} \cong \frac{R}{Rd_1} \oplus \cdots \oplus \frac{R}{Rd_k} \oplus \underbrace{\frac{R}{0} \oplus \cdots \oplus \frac{R}{0}}_{n-k}.$$

Since $R/0 \cong R$, the last $n-k$ terms give $R^{n-k}$. If some $d_i$ is a unit, then $Rd_i = R$ and $R/Rd_i = 0$, so that summand vanishes. The unit entries occur first in the divisibility chain; after deleting them and renaming the remaining nonzero non-unit entries $a_1, \ldots, a_r$, we retain $a_1 \mid \cdots \mid a_r$. Setting $s = n - k$, we obtain

$$M \cong R^s \oplus R/Ra_1 \oplus \cdots \oplus R/Ra_r. \qquad \square$$

**Definition 26** (9.2 — Torsion element)**.** *Let $R$ be a principal ideal domain and $M$ an $R$-module. An element $x \in M$ is called a* torsion element *if there exists a non-zero $r \in R$ such that $rx = 0$.*

**Definition 27** (9.4 — Torsion-free element)**.** *Let $R$ be a principal ideal domain and $M$ an $R$-module. A non-zero element $x \in M$ is called a* torsion-free element *if $rx = 0$ for $r \in R$ implies $r = 0$.*

**Theorem 52** (9.6)**.** *Let $R$ be a principal ideal domain and $M$ an $R$-module. Then*

$$\operatorname{Tor} M := \{x \in M \mid x \text{ is a torsion element}\}$$

*is a submodule of $M$.*

*Proof.* We verify the three conditions of Definition 4.8.

*Non-empty.* For any non-zero $r \in R$, $r \cdot 0 = 0$, so $0 \in \operatorname{Tor} M$.

*Closed under subtraction.* Let $x, y \in \operatorname{Tor} M$, say $rx = 0$ and $sy = 0$ for non-zero $r, s \in R$. Then

$$rs(x - y) = s(rx) - r(sy) = 0.$$

Since $R$ is a PID it is an integral domain, so $rs \neq 0$, hence $x - y \in \operatorname{Tor} M$.

*Closed under scalar multiplication.* Let $x \in \operatorname{Tor} M$ with $rx = 0$ for some non-zero $r \in R$, and let $a \in R$. Then $r(ax) = a(rx) = 0$, so $ax \in \operatorname{Tor} M$. $\square$

**Theorem 53** (9.7)**.** *Let $M$ be a finitely generated module over a principal ideal domain $R$. Then*

$$M = F \oplus \operatorname{Tor} M,$$

*where $F \cong R^s$ for some non-negative integer $s$, and $\operatorname{Tor} M \cong R/Ra_1 \oplus \cdots \oplus R/Ra_r$ with the $a_i$ non-zero non-units satisfying $a_1 \mid \cdots \mid a_r$.*

*Proof.* By Theorem 9.1, $M \cong R^s \oplus R/Ra_1 \oplus \cdots \oplus R/Ra_r$. Set $F \cong R^s$ and $T \cong R/Ra_1 \oplus \cdots \oplus R/Ra_r$, so $M = F \oplus T$. We show $T = \operatorname{Tor} M$.

$T \subseteq \operatorname{Tor} M$. We say $c \in R$ *annihilates* a module $N$ if $cy = 0$ for every $y \in N$, i.e. multiplying by $c$ sends everything in $N$ to zero. We show that the single non-zero element $a_r$ annihilates $T$, which makes every element of $T$ torsion.

*Step 1: $a_r \in Ra_i$ for each $i$.* Since $a_1 \mid a_2 \mid \cdots \mid a_r$, each $a_i$ divides $a_r$, i.e. $a_r = q_i a_i$ for some $q_i \in R$. Hence $a_r \in Ra_i$.

*Step 2: $a_r$ annihilates each summand $R/Ra_i$.* An element of $R/Ra_i$ is a coset $y + Ra_i$ with $y \in R$, and

$$a_r(y + Ra_i) = a_r y + Ra_i = yq_i a_i + Ra_i = Ra_i,$$

since $yq_i a_i \in Ra_i$ ($R$ is commutative). The coset $Ra_i$ is the zero element of $R/Ra_i$, so $a_r(y + Ra_i) = 0$.

*Step 3: $a_r$ annihilates $T$.* An element of $T$ is a tuple $t = (t_1, \ldots, t_r)$ with $t_i \in R/Ra_i$. Scalar multiplication acts coordinatewise, so by Step 2

$$a_r t = (a_r t_1, \ldots, a_r t_r) = (0, \ldots, 0) = 0.$$

Since $a_r \neq 0$, every $t \in T$ is a torsion element, so $T \subseteq \operatorname{Tor} M$.

$\operatorname{Tor} M \subseteq T$. Let $x \in \operatorname{Tor} M$ and write $x = x_1 + x_2$ with $x_1 \in F$ and $x_2 \in T$. Since $T \subseteq \operatorname{Tor} M$, the element $x_2$ is torsion, and therefore so is $x_1 = x - x_2$. Say $rx_1 = 0$ for some non-zero $r \in R$. Under the isomorphism $F \cong R^s$, $x_1$ corresponds to some $(y_1, \ldots, y_s) \in R^s$, and $r(y_1, \ldots, y_s) = 0$ gives $ry_i = 0$ for each $i$. Since $R$ is an integral domain and $r \neq 0$, each $y_i = 0$, so $x_1 = 0$. Thus $x = x_2 \in T$. □

**Theorem 54** (9.8)**.** *Let $R$ be a principal ideal domain. Then any non-zero ideal $P \neq R$ is prime if and only if it is maximal.*

*Proof.* ($\Leftarrow$) Every maximal ideal is prime by Theorem 6.21.

($\Rightarrow$) Let $P = (a)$ be a non-zero prime ideal. Suppose for a contradiction that $P$ is not maximal, so there exists an ideal $M$ with $P \subsetneq M \subsetneq R$. Since $R$ is a PID, $M = (b)$ for some $b \in R$. Then $(a) \subseteq (b)$ implies $a = bx$ for some $x \in R$. Since $b \notin (a)$ and $(a)$ is prime, the relation $a = bx \in (a)$ forces $x \in (a)$, say $x = ay$ for some $y \in R$. Then $a = bx = bay$, and since $R$ is an integral domain and $a \neq 0$, we may cancel to get $1 = by$. But then $b$ is a unit, so $M = (b) = R$, a contradiction. □

**Corollary 8.** *Let $R$ be a principal ideal domain and $P$ a non-zero prime ideal. Then $R/P$ is a field.*

*Proof.* Since $P$ is a non-zero prime ideal in the PID $R$, Theorem 9.8 gives that $P$ is maximal. It remains to show $R/P$ is a field, i.e. that every non-zero element $a+P \in R/P$ is invertible. Since $a \notin P$, the ideal $Ra+P$ strictly contains $P$. As $P$ is maximal, $Ra+P = R$, so there exist $r \in R$ and $p \in P$ with $ra+p = 1$. Then $(r+P)(a+P) = ra+P = 1+P$, so $a + P$ is invertible. □

**Theorem 55** (9.9 — Uniqueness of Decomposition)**.** *Let $R$ be a principal ideal domain and $M$ a finitely generated $R$-module. Suppose*

$$M \cong R^s \oplus R/Ra_1 \oplus \cdots \oplus R/Ra_u \quad \text{and} \quad M \cong R^t \oplus R/Rb_1 \oplus \cdots \oplus R/Rb_v,$$

*where the $a_i$ and $b_j$ are non-zero non-units with $a_1 \mid \cdots \mid a_u$ and $b_1 \mid \cdots \mid b_v$. Then $s = t$, $u = v$, and $Ra_i = Rb_i$ for all $i$.*

*Proof. Step 1: $s = t$.* By Theorem 9.7, both decompositions yield $M = F \oplus \operatorname{Tor} M$ with $F \cong R^s$ and $F' \cong R^t$ respectively. Then

$$R^s \cong F \cong M/\operatorname{Tor} M \cong F' \cong R^t,$$

so $s = t$ by Theorem 5.21. Let $T = \operatorname{Tor} M$; it remains to show $u = v$ and $Ra_i = Rb_i$.

*Step 2: $u = v$.* For an $R$-module $X$ and a prime $p \in R$, define $X_p := \{x \in X \mid px = 0\}$. By the corollary to Theorem 9.8, $R/Rp$ is a field, so $T_p$ is a vector space over $R/Rp$. From the first decomposition of $T$,

$$\left(\frac{R}{Ra_i}\right)_p = \begin{cases} R/Rp & \text{if } p \mid a_i, \\ 0 & \text{if } p \nmid a_i, \end{cases}$$

so $\dim_{R/Rp}(T_p)$ equals the number of $a_i$ divisible by $p$. Now choose a prime $p$ dividing $a_1$; since $a_1 \mid a_i$ for all $i$, $p$ divides every $a_i$, giving $\dim_{R/Rp}(T_p) = u$. Applying the same reasoning to the second decomposition, the dimension also equals the number of $b_j$ divisible by $p$, which is at most $v$, so $u \leq v$. By symmetry (choosing a prime dividing $b_1$), $v \leq u$, hence $u = v$.

*Step 3: $Ra_u = Rb_u$.* Since $a_i \mid a_u$ for all $i$, the element $a_u$ annihilates every summand in the first decomposition, so $a_u T = 0$. Applied to the second decomposition, $a_u \cdot (R/Rb_u) = 0$, which means $a_u \in Rb_u$, so $Ra_u \subseteq Rb_u$. By symmetry $Rb_u \subseteq Ra_u$, hence $Ra_u = Rb_u$.

*The remainder of the proof is non-examinable.* It remains to show $Ra_i = Rb_i$ for $i < u$ by downward induction on $i$. The idea is: assuming $Ra_i = Rb_i$ for $k \leq i \leq u$, one multiplies both decompositions of $T$ by a carefully chosen element $x \in R$ built from a prime $p$ and the exact $p$-adic valuation of $a_u$. This zeros out the first $k-1$ summands in one decomposition but not the other, producing two decompositions of $xT$ with a different number of non-zero summands — contradicting the fact (established by the dimension argument in Step 2) that any two such decompositions must have equally many non-zero summands. □

**Theorem 56** (9.10 — Fundamental Theorem of Finitely Generated Abelian Groups)**.** *Let $A$ be a finitely generated abelian group. Then*

$$A \cong \mathbb{Z}^s \oplus \mathbb{Z}/a_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/a_r\mathbb{Z},$$

*where $s \geq 0$, the $a_i$ are non-zero non-units in $\mathbb{Z}$, and $a_1 \mid a_2 \mid \cdots \mid a_r$. This decomposition is unique.*

*Proof.* Every abelian group is a $\mathbb{Z}$-module (with $n \cdot a$ meaning $a + \cdots + a$ or its inverse), and $\mathbb{Z}$ is a PID. So Theorem 9.1 applies directly with $R = \mathbb{Z}$, giving a decomposition $A \cong \mathbb{Z}^s \oplus \mathbb{Z}/a_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/a_r\mathbb{Z}$ where each $\mathbb{Z}/a_i\mathbb{Z} = R/Ra_i$ is a cyclic torsion summand. Uniqueness of $s$ and of the ideals $\mathbb{Z}a_i$ (hence the $a_i$ up to sign) then follows from Theorem 9.9.

If $A$ is finite, then $s = 0$: the summand $\mathbb{Z}^s$ contains elements of infinite order whenever $s \geq 1$, but a finite group has no such elements. □

# Lecture 13 — Rational Canonical Form

**Definition 28** (The $F[x]$-module structure on $V$)**.** *Let $F$ be a field, $V$ a finite-dimensional vector space over $F$, and $T \in \operatorname{Hom}_F(V, V)$ a linear map with matrix $A$ relative to some*

*basis. We make $V$ into an $F[x]$-module by defining, for any polynomial $f(x) = a_0 + a_1 x + \cdots + a_m x^m \in F[x]$ and any $v \in V$,*

$$f(x) \cdot v := a_0 v + a_1(Tv) + \cdots + a_m(T^m v).$$

*This extends the $F$-action on $V$ so that $x \cdot v = Tv$. Since any basis $\{v_1, \ldots, v_n\}$ of $V$ over $F$ also generates $V$ as an $F[x]$-module, $V$ is a finitely generated $F[x]$-module. Since $F[x]$ is a PID (as $F$ is a field), the structure theorem of Lecture 12 applies.*

**Theorem 57** (10.1)**.** *Let $T \in \mathrm{Hom}_F(V, V)$, and let $f_1(x), \ldots, f_n(x)$ be the invariant factors of $A - xI$, where $A$ is a matrix of $T$. Then*

$$V \cong \frac{F[x]}{(f_1(x))} \oplus \cdots \oplus \frac{F[x]}{(f_n(x))}.$$

*(Proof non-examinable.)* The key steps are: define the surjective $F[x]$-homomorphism $\phi : F[x]^n \to V$ by $e_i \mapsto v_i$, then show $\ker \phi = (A - xI)F[x]^n$, giving $V \cong F[x]^n/(A - xI)F[x]^n$. Since invertible changes of basis preserve this quotient up to isomorphism, one may apply Smith Normal Form (Theorem 8.8) to put $A - xI$ in diagonal form $\mathrm{diag}(f_1(x), \ldots, f_n(x))$. The image of $F[x]^n$ under this diagonal matrix is $(f_1(x)) \oplus \cdots \oplus (f_n(x))$, so Theorem 4.33 with $M_i = F[x]$ and $A_i = (f_i(x))$ for every $i$ gives

$$V \cong \frac{F[x] \oplus \cdots \oplus F[x]}{(f_1(x)) \oplus \cdots \oplus (f_n(x))} \cong \frac{F[x]}{(f_1(x))} \oplus \cdots \oplus \frac{F[x]}{(f_n(x))}.$$

**Remark 2** (10.2)**.** *The polynomials $f_1(x), \ldots, f_n(x)$ are called the* invariant factors *of $A$ (or of $T$). After discarding any unit factors, the remaining non-unit invariant factors are called the invariant factors of the $F[x]$-module $V$.*

**Definition 29** (10.3 — $T$-cyclic space)**.** *Let $T \in \mathrm{Hom}_F(V, V)$. If $V$, regarded as an $F[x]$-module via $T$, is cyclic, then $V$ is called a* $T$-cyclic space *and $T$ is called a* cyclic endomorphism *of $V$.*

**Theorem 58** (10.4)**.** *Let $W$ be a submodule of the $F[x]$-module $V$, and let $T \in \mathrm{Hom}_F(V, V)$ satisfy $TW \subseteq W$. Then $W$ is a $T$-cyclic subspace if and only if there exists $w \in W$ such that $\{w, Tw, \ldots, T^{k-1}w\}$ is a basis of $W$ for some $k \in \mathbb{N}$.*

*Proof.* ($\Rightarrow$) Suppose $W$ is $T$-cyclic, generated by $w$, so $W = F[x] \cdot w$ in the sense of Definition 4.13. Since $\dim V = n$ is finite, the vectors $w, Tw, T^2w, \ldots, T^n w$ must be linearly dependent over $F$. Let $T^r w$ be the first of these linearly dependent on the preceding ones, so $\{w, Tw, \ldots, T^{r-1}w\}$ are linearly independent and

$$T^r w = a_0 w + a_1(Tw) + \cdots + a_{r-1}(T^{r-1}w), \quad a_i \in F.$$

We claim every $T^i w$ is a linear combination of $\{w, Tw, \ldots, T^{r-1}w\}$. This is clear for $i \leq r-1$ and holds for $i = r$ by the equation above. For $i > r$, apply $T^{i-r}$ to the equation to get

$$T^i w = a_0(T^{i-r}w) + \cdots + a_{r-1}(T^{i-1}w),$$

and each term on the right is expressible in terms of $\{w, \ldots, T^{r-1}w\}$ by the induction hypothesis. Since $W = F[x] \cdot w$, every element of $W$ is of the form

$$f(x) \cdot w = (b_0 + b_1 x + \cdots + b_m x^m) \cdot w = b_0 w + b_1(Tw) + \cdots + b_m(T^m w),$$

using the $F[x]$-module action from the definition. Since every $T^i w$ lies in $\text{span}\{w, \ldots, T^{r-1}w\}$, this set spans $W$. Combined with linear independence, $\{w, Tw, \ldots, T^{r-1}w\}$ is a basis of $W$ with $k = r$.

($\Leftarrow$) Suppose $\{w, Tw, \ldots, T^{k-1}w\}$ is a basis of $W$. Any element $c_0 w + c_1(Tw) + \cdots + c_{k-1}(T^{k-1}w)$ of $W$ equals $(c_0 + c_1 x + \cdots + c_{k-1}x^{k-1}) \cdot w \in F[x] \cdot w$ by the module action, so $W \subseteq F[x] \cdot w$. Since $w \in W$ and $W$ is closed under the $F[x]$-action, $F[x] \cdot w \subseteq W$. Hence $W = F[x] \cdot w = (w)$, so $W$ is $T$-cyclic by Definition 10.3. $\square$

**Definition 30** (10.5 — Companion matrix)**.** *Let $f(x) = a_0 + a_1 x + \cdots + a_{k-1}x^{k-1} + x^k$ be a monic polynomial over $F$. The* companion matrix *of $f(x)$ is the $k \times k$ matrix*

$$C(f) = \begin{pmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_{k-1} \end{pmatrix}.$$

**Definition 31** (10.7 — Minimal polynomial of $T$ on $W$)**.** *Let $W$ be a $T$-cyclic subspace generated by $w$. Since $F[x]$ is a PID, the annihilator* $\text{Ann}(w) = \{f(x) \in F[x] \mid f(x) \cdot w = 0\}$ *is a principal ideal $(f(x))$ for a unique monic generator $f(x)$. This $f(x)$ is called the* minimal polynomial *of $T$ on $W$.*

**Theorem 59** (10.6)**.** *Let $T \in \text{Hom}_F(V, V)$ and let $W$ be a $T$-cyclic subspace of $V$ with basis $\{w, Tw, \ldots, T^{k-1}w\}$. Then the matrix of $T$ on $W$ is the companion matrix of the minimal polynomial $f(x)$ of $T$ on $W$.*

*Proof.* Since $\{w, Tw, \ldots, T^{k-1}w\}$ is a basis of $W$, the vector $T^k w$ is a linear combination of these basis elements, say

$$T^k w = a_0 w + a_1(Tw) + \cdots + a_{k-1}(T^{k-1}w), \quad a_i \in F.$$

In terms of the $F[x]$-module action ($x \cdot v = Tv$), this says $(a_0 + a_1 x + \cdots + a_{k-1}x^{k-1} - x^k) \cdot w = 0$, so the polynomial $g(x) := a_0 + a_1 x + \cdots + a_{k-1}x^{k-1} - x^k$ lies in $\text{Ann}(w) = (f(x))$.

Since $F[x]$ is a PID, $\text{Ann}(w)$ is a principal ideal $(f(x))$ for some monic $f(x)$ of degree $s$. The inclusion $g(x) \in (f(x))$ gives $s \leq k$. On the other hand, $f(x) \cdot w = 0$ means $T^s w$ is a linear combination of $\{w, Tw, \ldots, T^{s-1}w\}$. Since $\{w, Tw, \ldots, T^{k-1}w\}$ are linearly independent, this forces $s \geq k$. Hence $s = k$.

Since $g(x) \in (f(x))$, write $g(x) = c \cdot f(x)$ for some $c \in F$. Both $g$ and $f$ are monic of degree $k$ with leading coefficient $-1$ and 1 respectively, so $c = -1$ and $f(x) = -g(x) = x^k - a_{k-1}x^{k-1} - \cdots - a_1 x - a_0$.

The matrix of $T$ on $W$ relative to $\{w, Tw, \ldots, T^{k-1}w\}$ has columns given by the images of each basis vector: $T(T^i w) = T^{i+1}w$ for $i < k - 1$, contributing 1s on the subdiagonal, and the last column records the coefficients $a_0, \ldots, a_{k-1}$ of $T^k w$. By Definition 10.5, this is exactly the companion matrix of $f(x)$. $\square$

# Lecture 14

**Theorem 60** (10.8 — Rational Canonical Form)**.** *Let $V$ be a finite-dimensional vector space over a field $F$ and let $T \in \text{Hom}_F(V, V)$. Then there exists a basis of $V$, as an*

*$F[x]$-module, with respect to which the matrix of $T$ is*

$$\begin{pmatrix} B_1 & & \\ & \ddots & \\ & & B_r \end{pmatrix},$$

*where $B_i$ is the companion matrix of a monic polynomial $f_i(x)$, for $i = 1, \ldots, r$, such that $f_1(x) \mid f_2(x) \mid \cdots \mid f_r(x)$. This matrix is called the* rational canonical form *of $T$, and it is unique.*

*Proof. Step 1: $V$ is a torsion $F[x]$-module.* Since $\dim_F V = n$, we have $\dim_F \operatorname{Hom}_F(V, V) = n^2$. So the $n^2 + 1$ elements $I, T, T^2, \ldots, T^{n^2}$ must be linearly dependent over $F$. Let $T^k$ be the first element linearly dependent on the preceding ones, so

$$T^k = a_0 I + a_1 T + \cdots + a_{k-1} T^{k-1}, \quad a_i \in F.$$

Applying both sides to any $v \in V$ and using the $F[x]$-module action $x \cdot v = Tv$ gives

$$(a_0 + a_1 x + \cdots + a_{k-1} x^{k-1} - x^k) \cdot v = 0.$$

Since this polynomial is nonzero, every $v \in V$ is a torsion element (Definition 9.2), so $V = \operatorname{Tor} V$.

*Step 2: Decompose $V$ using Theorem 9.1.* $V$ is a finitely generated $F[x]$-module: any $F$-basis of $V$ also generates $V$ as an $F[x]$-module, since $F[x]$ is a PID. Since $V = \operatorname{Tor} V$, the free part in Theorem 9.1 vanishes, giving

$$V = V_1 \oplus \cdots \oplus V_r,$$

where each $V_i \cong F[x]/(f_i(x))$ is a cyclic $F[x]$-module and $f_1(x) \mid f_2(x) \mid \cdots \mid f_r(x)$.

*Step 3: Represent $T$ on each $V_i$.* Each $V_i$ is cyclic as an $F[x]$-module, hence a $T$-cyclic subspace (Definition 10.3). By Theorem 10.6, there exists a basis $\mathcal{B}_i$ of $V_i$ with respect to which $T|_{V_i}$ has matrix $B_i$, the companion matrix of $f_i(x)$.

*Step 4: Assemble.* Since $V = V_1 \oplus \cdots \oplus V_r$, the union $\mathcal{B} = \mathcal{B}_1 \cup \cdots \cup \mathcal{B}_r$ is a basis of $V$, and the matrix of $T$ with respect to $\mathcal{B}$ is $\operatorname{diag}(B_1, \ldots, B_r)$.

*Uniqueness.* The uniqueness of the decomposition subject to the divisibility chain, established in Theorem 9.9, implies the polynomials $f_i(x)$ — and hence the companion matrices $B_i$ — are uniquely determined by $T$. □

**Definition 32** (11.1 — Bi-additive map)**.** *Let $M$, $N$, and $P$ be abelian groups. A map $f : M \times N \to P$ is called* bi-additive *if*

$$f(m_1 + m_2,\, n) = f(m_1, n) + f(m_2, n) \quad \textit{and} \quad f(m,\, n_1 + n_2) = f(m, n_1) + f(m, n_2)$$

*for all $m, m_1, m_2 \in M$ and $n, n_1, n_2 \in N$.*

**Definition 33** (11.3 — Balanced map)**.** *Let $M$ be a right $R$-module and $N$ a left $R$-module. A map $f : M \times N \to P$ is called* balanced *if it is bi-additive and satisfies*

$$f(mr, n) = f(m, rn)$$

*for all $m \in M$, $n \in N$, and $r \in R$.*

**Definition 34** (11.5 — Tensor product)**.** *A* tensor product *of a right $R$-module $M$ and a left $R$-module $N$ is a pair $(T, \phi)$, where $T$ is an abelian group and $\phi : M \times N \to T$ is a balanced map, satisfying the following universal property: for every abelian group $P$ and every balanced map $f : M \times N \to P$, there exists a unique group homomorphism $\bar{f} : T \to P$ such that $\bar{f} \circ \phi = f$.*

**Remark 3** (11.6 — Uniqueness of the tensor product)**.** *If $(T', \phi')$ is another tensor product of $M$ and $N$, then there exists an isomorphism $\alpha : T \to T'$ such that $\phi' = \alpha \circ \phi$. Thus the tensor product is unique up to isomorphism. We write $M \otimes_R N$ for the tensor product and $m \otimes n$ for $\phi(m, n)$.*

**Theorem 61** (11.2 — Existence and construction of $M \otimes_R N$)**.** *Let $M$ be a right $R$-module and $N$ a left $R$-module. The tensor product $M \otimes_R N$ exists. Concretely, it is realised as $T(M, N) = A(M, N)/B(M, N)$, where $A(M, N)$ is the free abelian group with basis $M \times N$, and $B(M, N)$ is the subgroup generated by all elements of the form*

$$\begin{aligned}&(m_1 + m_2, n) - (m_1, n) - (m_2, n),\\ &(m, n_1 + n_2) - (m, n_1) - (m, n_2),\\ &(mr, n) - (m, rn),\end{aligned}$$

*for all $m, m_1, m_2 \in M$, $n, n_1, n_2 \in N$, and $r \in R$.*

*(Proof non-examinable.)*

# Lecture 15

**Theorem 62** (11.7)**.** *Let $M$ and $N$ be right and left $R$-modules, respectively. Then for all $m, m_1, m_2 \in M$, $n, n_1, n_2 \in N$, and $r \in R$:*

*(i)* $(m_1 + m_2) \otimes n = m_1 \otimes n + m_2 \otimes n$,

*(ii)* $m \otimes (n_1 + n_2) = m \otimes n_1 + m \otimes n_2$,

*(iii)* $mr \otimes n = m \otimes rn$.

*(Proof non-examinable.)* These follow directly from the fact that $B(M, N)$ maps to zero under $\phi$, so the defining relations of $B(M, N)$ become identities in $M \otimes_R N$.

**Theorem 63** (11.8)**.** *Let $M$ and $N$ be right and left $R$-modules, respectively.*

*(a) For $\lambda \in \mathbb{Z}$, $m \in M$, $n \in N$: $(\lambda m) \otimes n = \lambda(m \otimes n) = m \otimes \lambda n$. In particular, $0 \otimes n = 0 = m \otimes 0$ and $(-m) \otimes n = -(m \otimes n) = m \otimes (-n)$.*

*(b) Every element of $M \otimes_R N$ is a finite sum $\sum_i m_i \otimes n_i$ for $m_i \in M$, $n_i \in N$.*

*(Proof non-examinable.)* For (a), fix $n$; the map $m \mapsto m \otimes n$ is a group homomorphism, so it commutes with $\lambda$-multiples. For (b), every element of $A(M, N)$ is a finite $\mathbb{Z}$-linear combination $\sum \lambda_i (m_i, n_i)$, and since $\lambda_i(m_i \otimes n_i) = (\lambda_i m_i) \otimes n_i$ by (a), every element of $M \otimes_R N$ is a finite sum of simple tensors.

**Remark 4** (Module structure on $M \otimes_R N$ — 11.3)**.** *If $R$ is commutative, then $M \otimes_R N$ can be given the structure of a right $R$-module by defining $(m \otimes n)r = mr \otimes n$. This is well defined because $R$ being commutative ensures the map $fr : (m, n) \mapsto mr \otimes n$ vanishes on $B(M, N)$.*

**Theorem 64** (11.9)**.** *Let $M$ be a right $R$-module. Then $M \otimes_R R \cong M$ as additive abelian groups. If $R$ is commutative, then $M \otimes_R R \cong M$ as $R$-modules.*

*(Proof non-examinable.)* The map $\bar{f} : M \otimes_R R \to M$ defined by $\bar{f}(m \otimes r) = mr$ is a group isomorphism, with inverse $g : m \mapsto m \otimes 1$.

**Theorem 65** (11.10)**.** *Let $M$, $N$, and $P$ be $R$-modules over a commutative ring $R$. Then*

*(i) $M \otimes_R N \cong N \otimes_R M$ as $R$-modules,*

*(ii) $(M \otimes_R N) \otimes_R P \cong M \otimes_R (N \otimes_R P)$ as $R$-modules.*

*(Proof non-examinable.)* For (i), the map $m \otimes n \mapsto n \otimes m$ induces an isomorphism $M \otimes_R N \to N \otimes_R M$; the inverse is constructed symmetrically. Part (ii) is analogous.

**Definition 35** (Tensor product of homomorphisms — §11.4)**.** *Let $f : M \to M'$ and $g : N \to N'$ be $R$-homomorphisms. The* tensor product *$f \otimes g : M \otimes_R N \to M' \otimes_R N'$ is the homomorphism defined by*

$$(f \otimes g)(m \otimes n) = f(m) \otimes g(n).$$

*When $R$ is commutative, $f \otimes g$ is an $R$-homomorphism.*

**Theorem 66** (11.12)**.** *Let $f : M \to M'$, $f' : M' \to M''$, $g : N \to N'$, $g' : N' \to N''$ be $R$-homomorphisms. Then*

$$(f' \otimes g')(f \otimes g) = f'f \otimes g'g.$$

*In particular, if $f$ and $g$ are isomorphisms then $f \otimes g$ is also an isomorphism.*

*(Proof non-examinable.)* Follows directly from the definition: $(f' \otimes g')(f \otimes g)(m \otimes n) = f'f(m) \otimes g'g(n)$.

**Theorem 67** (11.13)**.** *Let $(M_i)_{i \in I}$ and $(N_j)_{j \in J}$ be families of right and left $R$-modules, respectively. Then*

$$\left( \bigoplus_{i \in I} M_i \right) \otimes_R \left( \bigoplus_{j \in J} N_j \right) \cong \bigoplus_{(i,j) \in I \times J} (M_i \otimes_R N_j),$$

*via the isomorphism $(\sum_i x_i) \otimes \left( \sum_j y_j \right) \mapsto \sum_{i,j} x_i \otimes y_j$.*

*(Proof non-examinable.)* The map $f$ sending $(\sum_i x_i, \sum_j y_j) \mapsto \sum_{i,j} x_i \otimes y_j$ is balanced, inducing a homomorphism $\bar{f}$ in one direction. The inclusions $\lambda_i : M_i \hookrightarrow \bigoplus M_k$ and $\mu_j : N_j \hookrightarrow \bigoplus N_\ell$ give homomorphisms $\lambda_i \otimes \mu_j$, which extend to an inverse $\bar{g}$; then $\bar{f}$ and $\bar{g}$ are mutually inverse isomorphisms.

**Theorem 68** (11.14)**.** *Let $R$ be a commutative ring, $f : M \to M'$ and $g : N \to N'$ be $R$-homomorphisms. If $f$ and $g$ are surjective, then $f \otimes g$ is surjective.*

*(Proof non-examinable.)* Every simple tensor $m' \otimes n' \in M' \otimes_R N'$ can be written as $f(m) \otimes g(n) = (f \otimes g)(m \otimes n)$ since $f$ and $g$ are surjective; hence every generator of $M' \otimes_R N'$ is in the image of $f \otimes g$.